

RELATIVE ARCHITECTURE · THE PROTOCOL · v1.1

The Language of Hierarchical Systems

Space. Rules. Dynamics. Diagnostics.

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PART I

THE SYSTEM

1 Chapter 1. What Is a Hierarchical System

A hierarchical system is a system in which elements are organized by levels of access, influence, or decision-making. Not all elements are equal. Not all connections are symmetric. Not all actions are available from every level.

This is not a definition by formula. It is a description of a property observed everywhere.

A company is a hierarchical system. An individual contributor, a department head, a commercial director, an owner — these are levels. Between them there is distance. This distance is not physical. It is determined by access to information, authority to make decisions, and ability to influence others.

A city is a hierarchical system. A factory, a municipal administration, small businesses, residents, cultural institutions — these are levels. Between them there is a structure of connections, and it is not flat.

A team is a hierarchical system. A product is a hierarchical system. A partnership between two organizations is a hierarchical system.

Wherever there are levels and unequal access, there is hierarchy.

1.1 Three Properties

Hierarchical systems possess three properties that distinguish them from flat structures. These properties are not metaphor. They describe observed behavior of systems. In mathematics and physics, they are formalized through ultrametric spaces (Vladimirov, Volovich, Zelenov, 1994; Rammal, Toulouse, Virasoro, 1986). The mathematical apparatus is not given here. The consequences are.

Property one: non-additivity.

The distance between two levels is not equal to the sum of small steps. One cannot reach the owner's level by visiting the immediate supervisor a hundred times. A transition is required — an event that opens access to another level. Small steps do not accumulate into a transition. They maintain the current state. An event transitions.

Property two: discreteness.

Between levels there are no intermediate states. Trust is either formal or expert. Access either exists or it does not. The system is either in one phase or another. One cannot be "70% a strategic partner." The scale describing hierarchical systems is discrete. The transition between levels is a jump, not a smooth movement.

Property three: the weak link determines the ceiling.

If a system has three elements and two are at a high level while one is at a low level, the system as a whole operates at the level of the low one. The lagging element does not pull down — it prevents rising higher. Until it is strengthened, the system will not move. This is not an arithmetic mean. This is a minimum that determines the maximum possible.

1.2 Consequences for Navigation

From the three properties follow three rules that operate throughout the entire model:

1. Small steps maintain. Events transition.
2. Levels are discrete. There are no intermediate values.
3. Priority goes to the lagging element. Not the strongest.

These rules are not recommendations. They are consequences of structure. They can be ignored, but the system will not change its properties because of that.

1.3 What Is Not a Hierarchical System

A flat network of equal elements is not a hierarchical system. In it, distances are additive, states are continuous, and the weak link does not determine the ceiling. This model does not apply to such systems.

In practice, purely flat networks almost do not exist. Any organization, any city, any product has at least minimal hierarchy. The model applies wherever there are at least two levels.

1.4 Boundaries of Applicability

This model describes the direction of system movement. It does not predict exact events. It does not say what will happen on a specific day. It says where the system will shift given a particular configuration and what transitions are possible.

This is not a limitation. This is a property of the level of description. Thermodynamics does not say where a molecule will fly. It says that gas will expand. This model says that upon accumulation of potential along certain axes, the system will transition to a new phase. Not when. Not through what event. But in what direction.

This is sufficient for navigation.

2 Chapter 2. Four Axes

Any hierarchical system possesses four independent parameters that define its state. These parameters are called axes. Each axis describes one aspect of the system. Together they define a space in which state can be recorded, dynamics tracked, and decisions made.

Four axes:

- **Z — Identity.** How the system understands itself.
- **Y — Quality.** What the system creates and embeds.
- **X — Connections.** How elements of the system are linked.
- **W — Resources.** Realized connection in exchange.

Each axis has three dimensions. Dimensions are numbered by a single rule:

Number	Aspect	Question
1	Topology	How is it structured?
2	State	What quality does it possess?
3	Dynamics	How does it move?

This rule is uniform across all four axes. It is not a convention. It follows from the fact that any system has structure, quality, and movement. Two aspects are insufficient. Four are redundant. Why exactly three and why in this order — Chapter 3.

2.1 Z — Identity

Axis Z describes how the system understands itself, its past, and its future. This is the level of narratives, language, and agency.

Three dimensions:

Dimension	Aspect	Description
Z ₁ . Language	Topology	Presence of a shared conceptual apparatus for describing processes, problems, and projects. How communication is structured.
Z ₂ . Agency	State	Who is perceived as the subject of change. Who sets the agenda and initiates action. What quality the role possesses.
Z ₃ . Vision	Dynamics	Degree of formation and shared acceptance of a long-term direction. Where the system is moving.

Z is a leading indicator. Changes along axis Z, as a rule, precede changes along the other axes. If the system does not understand where it is going, it can build neither quality, nor connections, nor resources.

2.2 Y — Quality

Axis Y describes what the system creates and how it is embedded in the environment. This is the level of integration, adaptation, and development.

Three dimensions:

Dimension	Aspect	Description
Y ₁ . Embeddedness	Topology	The degree to which what is created becomes part of everyday life rather than remaining a project. How it is structurally integrated into processes.
Y ₂ . Adaptation	State	The degree to which real needs of system elements are accounted for. What quality the fit possesses.
Y ₃ . Innovation	Dynamics	Emergence of new formats, practices, types of activity. How the system moves forward.

Y follows Z. Quality is created when there is understanding of for whom and for what. Without identity, quality remains a set of disconnected actions.

2.3 X — Connections

Axis X describes the structure of links between elements of the system. This is the level of breadth, depth, and density of interaction.

Three dimensions:

Dimension	Aspect	Description
X ₁ . Breadth	Topology	Number of independent centers and channels of connection. How the network is structured.
X ₂ . Depth	State	From formal protocol to joint decision-making. What quality trust possesses.
X ₃ . Density	Dynamics	Frequency and quality of interaction between elements. How the flow of contacts moves.

X is synchronous with Z. Connections expand and deepen as the system acquires a shared language and a shared vision. But X can also grow independently — through events that create new channels.

2.4 W — Resources

Axis W describes realized connection through resources — their volume, resilience, and temporal structure.

Three dimensions:

Dimension	Aspect	Description
W ₁ . Share	Topology	Volume of resources passing through non-central channels. How flows are structurally distributed.

W ₂ . Resilience	State	The ability of the system to withstand the loss of a key element. What quality the strength of connection possesses.
W ₃ . Duration	Dynamics	Temporal structure of commitments and projects. How it moves through time.

W is a lagging indicator. Resources change last. They are a projection of the state of Z, Y, and X, not an independent parameter of management.

2.5 Field Separation

The four axes are not equal in the nature of management. They separate into two fields.

Action field: Z, Y, X. These axes can be influenced directly. One can shape language and vision, create quality, build connections. One cannot guarantee the result, but one can guarantee the action. The response of the environment to actions replenishes knowledge about the system.

Result field: W. This axis cannot be influenced directly. Resources grow as a consequence of synchronization of Z, Y, and X. W is always a lagging indicator. It shows what has already happened, not what is happening.

From field separation follows a rule: do not manage W directly. Manage Z, Y, X. W will follow.

2.6 Asymmetry

Two scenarios of field relationship are possible.

Scenario one: Z, Y, X lead W. Potential is accumulated. Resources lag. This is normal for early stages of system development. No immediate action on W is required. Continued work on Z, Y, X is required.

Scenario two: W leads Z, Y, X. Resources exist. Identity, quality, and connections are not secured. This is an alarm signal. The connection holds on inertia or external compulsion, not on structure. At the first external disturbance, the system will lose resources, because they are not secured by coupling.

Confusion between these two scenarios is one of the principal sources of management errors.

2.7 Why Four

The four axes are not chosen arbitrarily. They satisfy three requirements.

Independence. Each axis can be advanced without changing the others. One can build connections without changing identity. One can create quality without increasing resources. The axes do not reduce to one another.

Necessity. If at least one axis equals zero, a stable system does not arise. Without identity there is no direction. Without quality there is no content. Without connections there is no structure. Without resources there is no realization. Four

are necessary. Three are insufficient. Five are redundant — a fifth axis always decomposes into a combination of four.

Operationality. Each axis can be measured through observable events. Not through opinions, not through feelings, but through facts: an event occurred or did not occur. If an event is recorded — the level is confirmed. If not — the level is not confirmed.

2.8 Why These

Axes Z, Y, X, W describe four fundamental questions about any system:

- **Z:** Where is the system going and who is it?
- **Y:** What does the system create?
- **X:** How are elements of the system connected?
- **W:** What does the system receive in exchange?

These questions do not depend on the domain. A city, a company, a team, a product, a partnership — any hierarchical system answers them. The content of the answers changes. The structure of the questions does not.

2.9 Summary Table

Axis	1 — Topology	2 — State	3 — Dynamics
Z — Identity	Z ₁ . Language	Z ₂ . Agency	Z ₃ . Vision
Y — Quality	Y ₁ . Embeddedness	Y ₂ . Adaptation	Y ₃ . Innovation
X — Connections	X ₁ . Breadth	X ₂ . Depth	X ₃ . Density
W — Resources	W ₁ . Share	W ₂ . Resilience	W ₃ . Duration

3 Chapter 3. Three Aspects

In Chapter 2 it was shown: each axis has three dimensions, and each dimension belongs to one of three aspects — topology, state, or dynamics. The numbering of dimensions follows a single rule: 1 = topology, 2 = state, 3 = dynamics.

Here it is explained why there are three aspects, why they are exactly these, and why exactly in this order.

3.1 Definitions

Topology — how it is structured. The aspect describes structure, form, configuration. Not quality and not movement, but arrangement: how many elements, how they are connected, what form they create.

The question of topology: “How is this structured?”

State — what quality it possesses. The aspect describes depth, maturity, strength. Not form and not speed, but quality: how strong, how deep, how mature.

The question of state: “What quality does this possess?”

Dynamics — how it moves. The aspect describes direction, frequency, development. Not form and not quality, but movement: where it goes, how fast, whether it reproduces.

The question of dynamics: “How does this move?”

3.2 Why Three

Any system — from a molecule to a city — has three fundamental properties: it is structured in some way, it possesses some quality, it moves in some way.

Two aspects are insufficient. Structure without quality is a schema without content. Quality without movement is a frozen snapshot. Movement without structure is chaos. Any pair leaves the system undescribed.

Four aspects do not exist. Any fourth aspect that can be proposed, upon examination, decomposes into a combination of three. “Function” is dynamics viewed through the prism of topology. “Potential” is state viewed in projection onto dynamics. “Scale” is topology viewed in projection onto state. A fourth aspect does not add a new dimension of description. It adds a new combination of existing ones.

Three aspects exhaust the description. This is not the author’s choice. This is a property of reality.

3.3 Why in This Order

The order $1 \rightarrow 2 \rightarrow 3$ is not arbitrary. It reflects the logical sequence of knowing a system.

First, structure is recorded: what exists, how it is connected, what form it creates. This is topology. Without it, one cannot speak of quality or movement, because it is unclear what exactly possesses quality and what exactly moves.

Then, quality is assessed: how strong, how deep, how mature. This is state. It has meaning only relative to already recorded structure. “Deep trust” is meaningless if it is unclear between whom and whom.

Then, movement is tracked: where it goes, how fast, whether it reproduces. This is dynamics. It has meaning only relative to structure and state. “Rapid development” is meaningless if it is unclear what develops and from what state.

The order: structure → quality → movement. From static to dynamic. From form to process.

3.4 How This Works in Diagnostics

Knowledge of the aspect of each dimension allows precise diagnoses.

If topology is high and dynamics is low — structure exists but it does not live. The network is built but contacts do not flow. Language exists but vision does not form. The system is frozen.

If dynamics is high and state is low — movement exists but it is empty. Contacts are frequent but trust is formal. Projects are launched but not embedded. The system bustles.

If state is high and topology is low — quality exists but it is not structured. Trust is deep but only with one person. Adaptation is precise but not embedded in processes. The system is fragile.

Without separation into aspects, these diagnoses are impossible. One can see that “axis X is at level 2.” But one cannot see that “the network is broad, trust is formal, the flow is active” — and these are three different problems requiring three different actions.

3.5 Connection with Phases

Aspects answer the question “what is measured.” Phase states answer the question “at what level.” These are two independent parameters of description.

For each dimension one can record: what aspect it describes (topology, state, or dynamics) and in what phase it is (0, 1, 2, 3, or 4). The aspect does not change. The phase changes.

The topology of a network can be at phase 1 (one channel) or at phase 4 (inter-team interaction). The aspect remains topology. The level changes.

This separation allows one to see the dynamics of a system without losing structure. A snapshot records phases. A radar shows how phases change. Aspects remain unchanged — they define the coordinate system within which movement occurs.

3.6 Summary Table

Axis	1 — Topology	2 — State	3 — Dynamics
Z — Identity	Z ₁ . Language	Z ₂ . Agency	Z ₃ . Vision

Y — Quality	Y ₁ . Embeddedness	Y ₂ . Adaptation	Y ₃ . Innovation
X — Connections	X ₁ . Breadth	X ₂ . Depth	X ₃ . Density
W — Resources	W ₁ . Share	W ₂ . Resilience	W ₃ . Duration

This table is identical to the summary table of Chapter 2. It is repeated here because Chapter 2 shows what, and Chapter 3 shows why. A reader who opens Chapter 3 without Chapter 2 must see the complete picture.

4 Chapter 4. Phase States

In Chapters 2 and 3 it was shown: any hierarchical system is described by four axes, each axis has three dimensions, and each dimension belongs to one of three aspects — topology, state, or dynamics.

Here the scale is described by which each dimension is assessed. The scale has five levels: 0, 1, 2, 3, 4. This is not an evaluation scale. These are phase states.

4.1 What Is a Phase

A phase is a qualitative state of a system, distinguished from other states not by degree but by nature. Water at 20°C and water at 80°C are the same phase: liquid. The difference is in degree. Water at 99°C and water at 101°C are different phases: liquid and gas. The difference is in nature.

Between phases there are no intermediate values. One cannot be “70% liquid and 30% gas.” The system is either in one phase or another. The transition between phases is a jump, not a smooth movement.

Levels 0–4 in this model describe phase states of a hierarchical system. Each level is a qualitatively different state. The transition between levels is an event, not an accumulation.

4.2 Five Phases

Level	Phase	Description
0	Absence	The system does not demonstrate this quality. No signs. No interaction.
1	Formal structure	Form appears. Interaction exists but is built on protocol, not on interest. Structure exists, movement does not.
2	Development	Fluidity appears. Interaction adapts, exchange becomes substantive. Form ceases to be rigid.
3	Stable state	Interaction fills the available space. Joint actions occur without external compulsion. The system reproduces itself.
4	Self-reproduction	The system generates quality independently. Does not depend on external impulse. New forms emerge organically.

4.3 Examples from Different Domains

Phases are universal. The same logic — absence, form, development, stability, self-reproduction — manifests in any hierarchical system. Below it is shown how phases appear in two domains unrelated to sales.

Domain: procurement. A buyer evaluates a supplier. Dimension X₂ — Depth.

Phase	How it manifests
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0	The supplier is not identified. No contact, no data.
1	The supplier is on the list. Interaction is strictly by procurement protocol. Request — response — delivery.
2	The buyer consults the supplier on specification. The supplier helps formulate the requirement.
3	The buyer defends the supplier before the internal customer. The supplier participates in demand planning.
4	Exchange of strategic information on procurement plans. Joint risk management. The supplier shapes the category agenda.

The transition from phase 1 to phase 2 is not achieved by ten emails with a commercial proposal. The transition requires an event: the supplier solves a problem the buyer could not solve alone. One event changes perception. Ten emails do not.

Domain: military. Interaction between units. Dimension X₂ — Depth.

Phase	How it manifests
0	Units do not know of each other's existence.
1	Interaction by regulation, through command. Connection is formal, procedural.
2	Joint exercises. Working out interaction. A shared language emerges.
3	Combat coordination. Mutual support without orders. Units act as a single whole.
4	Interchangeability. Joint operational planning. Units generate new tactics without external impulse.

The transition from phase 2 to phase 3 is not achieved by increasing the number of exercises. The transition requires an event: joint action in conditions where formal protocol does not work. One combat coordination changes trust in a way that twenty routine exercises do not.

4.4 Properties of the Phase Scale

Discreteness. Between levels there are no intermediate values. Trust is either formal or expert. Access either exists or it does not. One cannot be “at 2.5 on depth of trust.” If observable events confirm level 2 but do not confirm level 3 — the system is at level 2. Not “between 2 and 3.” At 2.

Non-additivity. The distance between levels is not equal to the sum of small steps. The transition from level 1 to level 2 is not achieved by ten touches at level 1. The transition requires an event comparable in scale to the distance between phases. Small steps maintain the current phase. Events transition to the next.

Irreversibility of transition. A phase transition, once made, does not reverse spontaneously. A system that has transitioned to level 3 will not return to level 1

without external destructive impact. This does not mean that a reverse transition is impossible — it is possible, but it requires its own event, not merely the cessation of effort.

4.5 Analogy with Phases of Matter

The analogy is not metaphorical. It is structural.

Level	Phase of matter	Property
0	Vacuum	No matter. No interaction.
1	Solid	Form exists. Rigid structure. No movement.
2	Liquid	Form changes. Fluidity appears. Adaptation.
3	Gas	Fills the available space. Free movement.
4	Plasma	Self-reproduction. The system generates energy itself.

Between solid and liquid there is no intermediate state. Between liquid and gas — none. Between gas and plasma — none. Likewise between levels 1 and 2, 2 and 3, 3 and 4 there are no intermediate values.

The transition from solid to liquid requires energy input to the critical point. Temperature rises, but the phase does not change. At the moment the critical point is reached, the phase changes in a jump. Likewise in a hierarchical system: small steps supply energy, but the phase does not change. An event reaches the critical point. The phase changes.

4.6 How This Works in Diagnostics

When assessing a dimension, one records the phase, not the degree. Not “how deep is trust” but “in what phase is trust.” Formal or expert. Expert or advocacy. There is no intermediate.

This removes subjectivity. The question “how much do you trust the supplier?” admits an infinite range of answers. The question “does the buyer consult the supplier on specification?” admits two answers: yes or no. If yes — level 2 is confirmed. If no — level 2 is not confirmed.

Presumption of unconfirmed: if the observable event corresponding to the criteria of a level is not recorded — the system remains at the current confirmed level. Not at the assumed one. At the confirmed one.

4.7 Connection with Ultrametrics and Spin Glasses

Discreteness of levels and non-additivity of transitions follow from the ultrametric nature of hierarchical spaces. In an ultrametric space, the distance between two points is determined not by the sum of steps but by the largest of them. Levels 0–4 are not a convenience marking. They are a description of states discovered in actual work with hierarchical systems. The mathematical apparatus is not given here. The consequence is recorded.

This ultrametric structure has a direct parallel in the physics of disordered systems, specifically in spin glass models (Sherrington–Kirkpatrick, p-spin). In these models, the system exists in an energy landscape with multiple metastable states separated by barriers. A transition between states requires overcoming a barrier and is not achieved by accumulating small fluctuations — which is structurally identical to the distinction between a small step and an event in this model.

Resources (axis W) correspond to the energy of the system in this analogy. Energy is a functional of the system’s configuration, not an independent control parameter. One cannot change energy without changing the configuration of spins; one cannot change W without synchronizing Z, Y, and X. The rule “do not manage W directly” thus receives confirmation in statistical physics: energy is a consequence, not a cause.

Desynchronization of axes ($\text{spread} \geq 2$) corresponds to frustration in spin glasses — a situation where interactions conflict and the system cannot reach a stable state without structural reconfiguration. Prioritizing the lagging dimension in this context is equivalent to relieving frustration: strengthening the weak link lowers the energy barrier for the transition of the entire system.

This analogy is not used as a calculation tool. It is provided as confirmation that the properties of hierarchical systems described in this protocol are not an arbitrary convention but reflect universal regularities of complex systems with ultrametric structure.

4.8 Summary Table

Level	Phase	Key property	Transition
0	Absence	No interaction	First event
1	Formal structure	Form exists, no movement	Event that creates interest
2	Development	Movement exists, form adapts	Event that changes perception
3	Stable state	Self-reproduction without external impulse	Event that changes role
4	Self-reproduction	The system generates quality itself	—

PART II

RULES

5 Chapter 5. Synchronization

In Part I it was shown: any hierarchical system is described by four axes, each axis has three dimensions, each dimension is in one of five phases. This gives a space of twelve dimensions in which the state of a system can be recorded.

But twelve numbers by themselves do not yield a diagnosis. One can have high values on some dimensions and low on others. One can have all axes at the same level but within one axis — a spread. One can have all dimensions synchronized but one axis lagging behind the rest.

To turn a set of numbers into a diagnosis, rules are needed. The first and principal of them is the rule of synchronization.

5.1 Two Rules

Rule 1: the spread within an axis does not exceed 1.

Each axis has three dimensions. Spread is the difference between the maximum and minimum value among the three dimensions. If the spread equals 0 or 1 — the axis is synchronized. If the spread equals 2 or more — internal desynchronization is recorded.

Example. Axis X: $X_1 = 3$, $X_2 = 3$, $X_3 = 2$. Spread = 1. The axis is synchronized. The network is broad, trust is deep, the flow is active. The system is stable.

Example. Axis X: $X_1 = 3$, $X_2 = 1$, $X_3 = 3$. Spread = 2. Internal desynchronization. The network is broad, the flow is active, but trust is formal. The system is unstable: at the first crisis, formal trust will not hold the connection.

Rule 2: the spread between axes does not exceed 1.

The four axes have mean values. The spread between axes is the difference between the maximum and minimum mean. If the spread equals 0 or 1 — the system is synchronized. If the spread equals 2 or more — inter-axis desynchronization is recorded.

Example. $Z = 3.00$, $Y = 2.67$, $X = 3.00$, $W = 2.33$. Spread = 0.67. The system is synchronized. All axes move in concert.

Example. $Z = 3.00$, $Y = 3.00$, $X = 3.00$, $W = 0.33$. Spread = 2.67. Inter-axis desynchronization. Identity, quality, and connections are at a high level. Resources are at zero. This is a growth zone: potential is accumulated but not realized.

5.2 What Is Desynchronization

Desynchronization is a state in which elements of a system move at different speeds or are at different levels. The system does not collapse, but it loses stability. The greater the spread, the higher the risk of cascading collapse.

Internal desynchronization is more dangerous than inter-axis desynchronization. If within an axis one dimension sharply lags, it determines the real level of the entire axis. Axis X with values 3, 1, 3 operates at level 1, not at level 2.33. The mean is misleading. The minimum determines reality.

This is a direct consequence of the weak link rule formulated in Chapter 1: the lagging element does not pull down — it prevents rising higher.

5.3 Priority to the Lagging Dimension

Upon discovery of desynchronization, priority is given to the lagging dimension. Not the strongest. Not the average. The lagging one.

This is not a recommendation. This is a consequence of structure. Until the lagging dimension is strengthened, the system will not move. Strengthening strong dimensions while the weak one lags yields no result — the ceiling is determined by the minimum.

Example. Axis Z: $Z_1 = 3$, $Z_2 = 1$, $Z_3 = 3$. Language and vision are at a high level. Agency is low. The system knows where it is going and has a shared language, but does not have the right to go there. Priority — Z_2 . Not Z_1 , not Z_3 . Z_2 .

Example. Inter-axis desynchronization: $Z = 3.00$, $Y = 3.00$, $X = 3.00$, $W = 0.33$. Priority — not W . W cannot be managed directly. Priority — maintaining Z , Y , X at the current level and waiting for W to follow. If W does not follow within several cycles — strategy revision.

5.4 Examples from Different Domains

Domain: M&A. Coupling of two organizations after a deal.

Snapshot: $X_1 = 3$, $X_2 = 1$, $X_3 = 3$. The network between teams is broad, the flow of interaction is active, but trust is formal. Teams communicate by integration protocol but do not trust each other. Spread = 2. Internal desynchronization. Risk: at the first conflict, formal trust will not hold the connection. Teams will revert to the “us — them” model. Priority: X_2 . Not X_1 , not X_3 . X_2 .

Snapshot: $Z = 3.00$, $Y = 3.00$, $X = 3.00$, $W = 0.33$. Identity, quality, and connections are at a high level. Teams speak a shared language, processes are integrated, trust is deep. But resources are not realized: synergies are on paper, not in the budget. Spread = 2.67. Inter-axis desynchronization. This is a growth zone: potential is accumulated but not converted. Priority: not W . Maintaining Z , Y , X . W will follow when integration is complete.

Snapshot: $Z_1 = 3$, $Z_2 = 1$, $Z_3 = 3$. Language is shared, vision is shared, but agency is low — one team dominates, the other executes. Spread = 2. Internal desynchronization. Risk: the dominating team loses interest, the subordinate team sabotages. Priority: Z_2 . Not Z_1 , not Z_3 . Z_2 .

Domain: urban management. Interaction between groups in a city.

Snapshot: $Z_1 = 3$, $Z_2 = 1$, $Z_3 = 3$. Language is shared, vision is shared, but agency is low — residents are perceived as objects of management, not subjects. Spread = 2. Internal desynchronization. Risk: upon change of administration or factory, the vision will collapse because it is not supported by agency. Priority: Z_2 . Not Z_1 , not Z_3 . Z_2 .

Snapshot: $Z = 2.00$, $Y = 3.00$, $X = 3.00$, $W = 3.00$. Quality, connections, and resources are at a high level. Projects are embedded, trust is deep, the economy is diversified. But identity lags: the city does not understand where it is going. Spread = 1.00. The system is synchronized, but Z lags. Priority: Z. Not Y, not X. Z.

Snapshot: $X_1 = 4$, $X_2 = 2$, $X_3 = 4$. The network is broad, the flow is continuous, but trust is formal. Groups communicate constantly but do not trust each other. Spread = 2. Internal desynchronization. Risk: at the first conflict, formal trust will not hold. Priority: X_2 . Not X_1 , not X_3 . X_2 .

5.5 Connection with Aspects

Knowledge of the aspect of each dimension allows one to refine the diagnosis of desynchronization.

If within an axis topology lags (dimension 1) — structure is not built. The network is narrow, embeddedness is low, share is small. Action: expand the structure.

If state lags (dimension 2) — quality is not achieved. Trust is formal, adaptation is weak, resilience is low. Action: deepen quality through an event.

If dynamics lags (dimension 3) — movement is absent. The flow is rare, there are no innovations, the horizon is short. Action: create an event that launches movement.

The aspect shows what exactly lags. The number shows the level. Together they give a complete diagnosis: not “axis X is at level 2” but “the network is broad, trust is formal, the flow is active.” These are three different problems requiring three different actions.

5.6 Summary Table of Rules

Rule	Condition	Status	Action
Spread within axis ≤ 1	$\max - \min \leq 1$	Synchronized	Maintenance
Spread within axis ≥ 2	$\max - \min \geq 2$	Internal desync.	Priority to lagging dimension
Spread between axes ≤ 1	$\max - \min \leq 1$	Synchronized	Maintenance
Spread between axes ≥ 2	$\max - \min \geq 2$	Inter-axis desync.	Priority to lagging axis
W lags Z, Y, X	$W < \min(Z, Y, X)$	Growth zone	Continued work on Z, Y, X
W leads Z, Y, X	$W > \max(Z, Y, X)$	Risk zone	Urgent actions on Z, Y, X

6 Chapter 6. Field Separation and Asymmetry

In Chapter 2 it was recorded: the four axes separate into two fields. The action field — Z, Y, X. The result field — W. In Chapter 5 it was shown how synchronization works within and between axes. Here it is described how the two fields interact over time and why they cannot be confused.

6.1 Why the Fields Are Separated

Axes Z, Y, and X respond to actions. One can shape language, create quality, build connections. One cannot guarantee the result, but one can guarantee the action. The response of the environment to actions replenishes knowledge about the system.

Axis W does not respond to actions directly. One cannot by a single precise intervention immediately increase the share of resources or make commitments long-term. W grows as a consequence of synchronization of Z, Y, and X.

This is not a convention. This is a property of hierarchical systems. Resources are a projection of the state of the system, not its parameter. The projection changes after the state has changed. Not simultaneously. Not before. After.

6.2 W as a Lagging Indicator

W always shows what has already happened. Not what is happening. Not what will happen.

If W is growing — it means that actions on Z, Y, X were correct some time ago. If W stands still while Z, Y, X are growing — this does not mean that actions are not working. This means that the delay has not yet passed. If W is falling while Z, Y, X are high — this is a signal: possibly the state on non-financial axes has been assessed incorrectly, or an external destructive impact has occurred that has not yet manifested in Z, Y, X.

The delay of W is not a defect of the model. This is a property of reality. Contracts require approvals. Budgets are drawn up by cycle. Decisions about resources are made at levels that do not see everyday interaction. W cannot react instantly because resource decisions are inert by nature.

6.3 Two Scenarios

From field separation and the lag of W follow two scenarios. Confusion between them is one of the principal sources of management errors.

Scenario one: Z, Y, X lead W. Growth zone.

Non-financial axes are at certain levels. W lags. Resources are not yet realized.

This is a normal state for early stages of system development. Potential is accumulated: language is formed, quality is created, connections are built. W will follow with a delay.

Action: do not press W. Continue work on Z, Y, X. Maintain synchronization. Wait for a window of opportunity to convert potential into resources.

Error: demanding immediate realization of resources. An attempt to force W destroys non-financial axes — trust is burned, quality is devalued, agency is lost.

Scenario two: W leads Z, Y, X. Risk zone.

Resources exist. Identity, quality, and connections are not secured. The system receives resources but does not have the structure that holds them.

This is an alarm signal. The connection holds on inertia, external compulsion, or accidental coincidence of circumstances. At the first external disturbance, the system will lose resources because they are not secured by coupling.

Action: urgent actions on Z, Y, X. Not on W. Pressing the system and demanding even more resources in such a situation means accelerating the rupture. One must deepen trust, raise quality, synchronize language.

Error: perceiving high W as a sign of well-being. Ignoring the lag of Z, Y, X. Believing that “if resources are flowing, everything is fine.”

6.4 The Rule

From the two scenarios follows a single rule:

Do not manage W directly. Manage Z, Y, X. W will follow.

This is not a recommendation. This is a consequence of structure. W is a projection. A projection cannot be changed without changing the object. One can demand growth of resources as much as one wants — if Z, Y, X are not synchronized, resources either will not come or will come and leave.

And conversely: if Z, Y, X are synchronized at a high level, W will follow inevitably. The question is only one of delay.

6.5 What Happens When the Rule Is Violated

The rule “do not manage W directly” is not a recommendation. Its violation leads to destruction of the system. Two typical scenarios.

Scenario one: forcing change.

The leadership of a company decides to implement a new organizational structure. W is forced: budgets are allocated, people are hired, tools are purchased. But Z, Y, X are not ready: language is not formed — divisions speak different languages and do not understand the goals of the reform. Quality is not embedded — new processes exist on paper but not in everyday life. Connections are formal — teams do not trust each other and interact by protocol.

Changes are “pushed through” by order. The structure changes formally. But at the first crisis — missed deadlines, conflict between divisions, change of leader — the system reverts to the old. Not because the structure was bad. But because it was not secured by language, quality, and connections. W was high, but Z, Y, X were empty. High W without structure is not change but a transaction. A transaction ends. Change does not.

Scenario two: flooding with resources.

A company faces a problem: staff turnover is growing, key specialists leave, teams do not hold results. Diagnostics show: the problem is in Z and X. Language is not shared — divisions do not understand the company's strategy. Vision is not shared — people do not know where the organization is going. Trust is formal — teams interact by protocol but do not support each other. Agency is low — employees execute but do not initiate.

Instead of working on Z and X, leadership floods W: raises salaries, introduces bonuses, expands the social package. W grows. Exchange proceeds. But exchange proceeds through an unprepared structure.

The weak link does not strengthen — it breaks under load. The higher the salaries, the higher the expectations. The higher the expectations, the stronger the disappointment at the first failure. If trust is formal and compensation grows, any conflict is perceived not as a working situation but as betrayal. If language is not shared and bonuses are tied to KPIs, any discrepancy in evaluation becomes a reason to leave. If agency is low and resources grow, employees perceive the company as an ATM, not as an environment.

The problem is not solved. It worsens. Because the load on the weak link grows, but its strength does not. People leave not because they are paid too little. They leave because the structure does not hold. And the structure does not hold because Z and X were not built.

General rule. W without Z, Y, X is exchange without structure. Exchange without structure does not create connection. It creates dependency, which is destroyed at the first external disturbance. One must manage Z, Y, X. W will follow. If W grows while Z, Y, X do not grow — this is not growth. This is accumulation of fragility.

6.6 Examples from Different Domains

Domain: M&A. Coupling of two organizations after a deal.

Growth zone scenario: $Z = 2.67$, $Y = 2.33$, $X = 2.67$, $W = 0.67$. Teams speak a shared language, processes begin to integrate, trust grows. But synergies are on paper, not in the budget. Spread = 2.00. Inter-axis desynchronization. This is a growth zone. Action: do not force W. Continue integration of teams and processes. W will follow when integration is complete.

Risk zone scenario: $Z = 1.00$, $Y = 1.33$, $X = 1.00$, $W = 3.00$. The deal is closed, resources are consolidated, budgets are merged. But teams do not know each other, processes are not integrated, language is different. Spread = 2.00. Inter-axis desynchronization. This is a risk zone. Action: urgent actions on Z, Y, X. Joint working groups, shared language, first joint projects. Do not demand synergies — they will not come without integration.

Domain: urban management. Interaction between groups in a city.

Growth zone scenario: $Z = 2.33$, $Y = 2.00$, $X = 2.00$, $W = 0.33$. Language is forming, projects appear, connections expand. But resources still flow through one

center. Spread = 2.00. Growth zone. Action: continue work on Z, Y, X. Resources will diversify when the system becomes a subject.

Risk zone scenario: $Z = 1.00$, $Y = 1.00$, $X = 1.00$, $W = 3.00$. Resources exist — the factory finances, grants come, projects are realized. But language is not formed, projects are not embedded, connections are formal. Spread = 2.00. Risk zone. Action: do not increase resources. Form language, embed projects, deepen trust. Otherwise, upon change of financing, the system will collapse.

6.7 Connection with Synchronization

Field separation and synchronization work together. Synchronization (Chapter 5) answers the question: are the axes coordinated with each other? Field separation answers the question: in what direction does the desynchronization work?

If desynchronization is in the direction “Z, Y, X lead W” — this is a growth zone. Potential is accumulated, realization is expected.

If desynchronization is in the direction “W leads Z, Y, X” — this is a risk zone. Resources are not secured by structure.

If desynchronization is within one of the axes Z, Y, X — this is an internal problem, solved by priority to the lagging dimension (Chapter 5).

If desynchronization is between W and the other axes — this is a problem of direction, solved by the rule: do not manage W directly.

6.8 Summary Table

Configuration	Status	Action	Error
$Z, Y, X > W$	Growth zone	Continued work on Z, Y, X	Forcing W
$W > Z, Y, X$	Risk zone	Urgent actions on Z, Y, X	Ignoring the lag
$Z, Y, X \approx W$	Synchronization	Maintenance	Complacency
$W = 0$ with $Z, Y, X > 0$	Early stage	Patience, continued work	Panic, pressure
W falls with Z, Y, X high	Alarm signal	Recheck Z, Y, X, search for external impact	Ignoring

7 Chapter 7. Structural Ceiling

In Chapters 5 and 6 the rules of synchronization and field separation were described. They answer the question: how to manage a system when axes can grow. Here a situation is described in which growth is impossible. Not because effort is insufficient. But because the environment does not allow it.

7.1 Definition

A structural ceiling is an objective constraint of the environment that does not allow an axis or dimension to grow above a certain level, regardless of effort applied.

This is not a temporary halt. This is not stagnation. This is a property of the system. Some systems cannot grow above a certain level — not because work is done poorly, but because their structure does not presuppose such growth.

Example. A small city with a population of 10 thousand people cannot become a federal center of attraction. Not because residents do not try. But because scale does not allow it. W_1 (share of external resources) will not grow above a certain value, no matter how much Z , Y , and X improve.

Example. A division of a company performing a support function cannot become a strategic partner for the head office. Not because the team works poorly. But because the role does not presuppose such a level of agency. Z_2 (agency) will not grow above a certain value, no matter how much trust deepens.

7.2 How to Distinguish a Ceiling from Stagnation

A structural ceiling and stagnation look the same: the axis does not grow. But their nature is different.

Stagnation is a halt in growth in the absence or insufficiency of effort. The axis does not grow because work has stopped or is being conducted incorrectly. Stagnation is cured by resuming work. If effort is resumed and the axis begins to grow — it was stagnation.

Structural ceiling is a halt in growth with sufficient and methodical effort. The axis does not grow because the environment does not allow it. A ceiling is not cured by resuming work. If effort is conducted methodically and the axis remains at the same level over several cycles — a ceiling has likely been reached.

Criterion of distinction: three cycles of methodical work without significant shifts while changing types of interventions and target dimensions. If after three cycles the axis has not shifted — this is not stagnation. This is a ceiling.

7.3 Internal Structural Ceiling

A structural ceiling is not always external. Often it is created by processes within the system.

Divisions may respond “impossible” to a non-standard request without examining the task. Leadership may not provide a platform for discussing complex

projects. Adjacent groups may not support an initiative if it goes beyond standard procedures.

All of these are internal barriers that block the growth of axes not because the environment does not allow it, but because the system itself is not ready for non-standard action.

Diagnostics of internal barriers is a separate task. It differs from analysis of external constraints: the source of the constraint is inside the system, not in the environment.

Example. A company attempts to build a strategic partnership with a key client. Z and X grow: language is synchronized, trust deepens. But Y does not grow: the offering is not adapted, there are no joint developments. The cause is not the client. The cause is internal: the engineering department does not allocate resources for customization, marketing is occupied with tactical tasks, leadership does not prioritize joint projects. Internal structural ceiling.

Example. A city attempts to diversify its economy. Z grows: language is forming, agency is distributing. But W does not grow: external resources do not come. The cause is not investors. The cause is internal: the administration does not create conditions for small business, regulatory barriers are not removed, infrastructure is not ready. Internal structural ceiling.

7.4 Two Strategies

When a ceiling is discovered, two strategies are possible.

Strategy one: alignment.

If one axis has reached a ceiling while others lag, effort is directed at the lagging axes. Let W not grow — but Z, Y, X can be strengthened. Let Z_2 not grow above 2 — but X_2 and Y_1 can be deepened.

Alignment does not increase the overall level of the system. But it increases stability. A system with synchronized axes at level 2 is more stable than a system with one axis at level 4 and the rest at level 1.

Example. A division of a company cannot become a strategic partner for the head office ($Z_2 = 2$, ceiling). But trust between teams can be deepened (X_2), the quality of joint processes improved (Y_1), language synchronized (Z_1). The system will not grow above level 2 on Z_2 , but will become stable at this level.

Strategy two: expansion.

If the environment allows, the system can be offered what it has not done before. Not to increase the level in the current domain, but to enter an adjacent one. Then the overall volume of interaction will grow, and the connection will become more multidimensional and durable.

Expansion does not remove the ceiling. It bypasses it. The ceiling remains, but the system finds new space for growth.

Example. A city cannot attract a major investor ($W_1 = 2$, ceiling). But it can develop tourism, small business, creative industries. W_1 will not grow, but the overall volume of resources will increase through new channels. The ceiling on one dimension is bypassed through growth on others.

Example. A division cannot become a strategic partner on the main product ($Y_2 = 2$, ceiling). But it can offer adjacent services: consulting, training, analytics. Y_2 will not grow, but the overall volume of interaction will increase. The ceiling is bypassed through expansion of the offering.

7.5 Ceiling as Signal

Discovery of a growth constraint on an axis is data about the environment. It shows where to direct resources and where not to direct them.

A structural ceiling is a signal. It shows the boundaries of the environment. A system that accounts for ceilings is more stable than a system that attempts to grow everywhere simultaneously.

7.6 Summary Table

Situation	Indicator	Strategy
Axis does not grow, effort has stopped	Stagnation	Resume work
Axis does not grow, effort continues, 1-2 cycles	Possible stagnation or ceiling	Change type of interventions
Axis does not grow, effort continues, 3+ cycles	Structural ceiling	Alignment or expansion
Ceiling is external	Environment does not allow	Expansion into adjacent domain
Ceiling is internal	Source of constraint is inside the system	Diagnostics of internal barriers
One axis at ceiling, others lag	Desynchronization	Alignment: priority to lagging

8 Chapter 8. Small Steps and Events

In Chapters 5–7 the rules of synchronization, field separation, and structural ceiling were described. They answer the question: how the space is structured and what constraints operate within it. Here it is described how to act within this space. Two types of interventions: small steps and events.

8.1 Small Steps

A small step is an intervention that maintains the system at the current level but does not transition it to the next.

Small steps do not move upward. This is not their function. Their function is to counteract entropy. Without an inflow of energy, the system degrades: information is forgotten, connections cool, people leave, newcomers do not know the history. Small steps are the energy supplied to the system so that it does not fall apart.

Examples of small steps: regular information sharing, a greeting, a short status check, a link to useful material, participation in a routine meeting. Each such action does not advance the system forward but does not allow it to degrade.

8.2 Principle of Proportional Support

Not all small steps are equal. What maintains the system at level 1 is insufficient for maintaining it at level 3.

At level 1 (formal structure), rare touches are sufficient: compliance with protocol, response to requests, keeping the channel open.

At level 2 (development), substantive occasions are needed: useful information, consultations, data exchange.

At level 3 (stable state), small steps must be even more substantial: joint problem-solving, exchange of market data, invitations to events.

At level 4 (self-reproduction), small steps become part of joint activity: strategic sessions, joint projects, informal meetings.

The principle: the minimum sufficient energy for maintaining a level grows with the level itself. The higher the system has risen, the more substantial everyday interventions must be.

This is a direct consequence of non-additivity described in Chapter 1. Distances between levels are not equal. Each subsequent level requires steps incommensurable with those that worked before. One cannot continue doing the same thing and expect the same result — because the space itself has changed.

8.3 Events

An event is an intervention that transitions the system to the next level. Does not maintain. Transitions.

An event differs from a small step not in scale but in nature. A small step operates within the current phase. An event changes the phase. A small step says: “I am here, remember me.” An event says: “Look at me differently.”

Three properties of an event:

First: independent value. The event carries value for the system regardless of whether the interaction will continue. Analytics that help make decisions. Expertise that reveals a hidden problem. Information that expands the picture. Value exists separately from the offering.

Second: absence of immediate request. The event does not require immediate reciprocal actions from the system. Not “let’s meet,” not “sign the contract,” not “allocate budget.” Something is given that works on its own. This reduces defensive reaction and increases the probability that the intervention will reach its target.

Third: shift in perception. After the event, the system begins to see the source of intervention differently. Not as a formal participant but as a carrier of valuable expertise. Not as an object of management but as a subject. Not as a replaceable element but as an irreplaceable one. It is this shift in perception that opens the door to the next level.

8.4 Why Small Steps Do Not Accumulate

In Chapter 1 it was recorded: the distance between levels is not equal to the sum of small steps. This is the property of non-additivity. Now it can be shown how it manifests in action.

Most standard actions — regular touches, template proposals, routine correspondence — are small steps. They maintain contact but do not change the level. One can send ten emails describing a proposal — trust will not become deeper from this. One can clarify needs twenty times — agency will not grow from this. One can hold a meeting by protocol a hundred times — joint decision-making will not arise from this.

This does not mean that small steps are useless. Their role is maintenance and preparation. But they cannot transition the system to the next level. For that, an event is required.

8.5 The Catalyst Paradox

There is an apparent contradiction. On one hand, small steps do not transition the system to the next level. On the other hand, sometimes a small step triggers a transition. How to reconcile this?

The resolution lies in distinguishing cause and catalyst. Small steps are not the cause of a transition. But they can become a catalyst if the system is already in a pre-critical state.

If potential along axes Z, Y, and X is already accumulated, levels are high, synchronization exists, but the transition has not yet occurred — a trigger is missing. At this moment, even a small action — a correctly posed question, timely provided information, a short meeting without an agenda — can trigger a cascade. The system, which was already ready for transition, receives the final impulse and restructures.

But the same small step, performed in a system where potential is not accumulated, will yield nothing. It will be just another touch. The effect depends not on the magnitude of the step but on the state of the system.

Criterion: if after a small step the system shifted a level — it was a catalyst in a pre-critical system. If it did not shift — it was an ordinary small step. The difference is not in the action. The difference is in the state of the environment.

8.6 The Rule

From all of the above follows a single rule:

Small steps maintain. Events transition.

This is not a recommendation. This is a consequence of structure. Small steps counteract entropy. Events change the phase. Confusing them means either spending resources on the impossible (trying to grow through small steps) or risking what has been accumulated (not maintaining the level between events).

Both types of interventions are necessary. Without small steps, the system degrades between events. Without events, the system remains at the current level, no matter how many small steps are made.

8.7 Examples from Different Domains

Domain: education / research school. Interaction between research groups.

Small step: regular seminars, preprint mailings, participation in conferences without joint presentations. Maintains awareness. Does not create joint work.

Event: two groups for the first time jointly publish a work that receives recognition in the community. Changes perception: “they” become “we.” Transitions X_2 from level 1 (formal acquaintance) to level 2 (substantive exchange).

Catalyst: if groups already know each other’s work, cite each other, meet at conferences, but lack the first joint experience — a short joint seminar without a formal agenda can trigger the transition. The same seminar in a system where groups do not know each other will yield nothing.

Domain: product / user. Interaction between a product team and users.

Small step: regular updates, mailings, responses to support requests. Maintains the current level of engagement. Does not create loyalty.

Event: a user for the first time independently proposes an improvement that the team accepts and implements without corrections. Changes perception: the user ceases to be a consumer and becomes a co-author. Transitions Z_2 from level 1 (user executes a scenario) to level 2 (user initiates).

Catalyst: if the user is already loyal, regularly gives feedback, but lacks the first accepted proposal — an open request for improvements without constraints can trigger the transition. The same request in a system where the user is not engaged will receive formal responses.

Domain: organizational management. Internal transformation of a company.

Small step: regular mailings about the progress of reform, status meetings, training by protocol. Maintains awareness. Does not create engagement.

Event: a division for the first time independently proposes a solution that changes a process, and leadership accepts it without corrections. Changes perception: employees cease to be executors. Transitions Z_2 from level 1 (execution) to level 2 (initiative).

Catalyst: if the language of reform is already understood, trust in leadership exists, but employees have not yet shown initiative — an open request for proposals without constraints can trigger the transition. The same request in a system where language is not formed will receive formal responses.

8.8 Summary Tables

Type of intervention	Function	Property	Result
Small step	Maintenance	Counteraction to entropy	System remains at current level
Event	Transition	Phase change	System transitions to next level
Catalyst	Trigger of transition	Small step in a pre-critical system	System transitions to next level

Property of an event	Description
Independent value	Value exists separately from the offering
Absence of immediate request	Does not require reciprocal actions
Shift in perception	System sees the source of intervention differently

9 Chapter 9. Entropy and Maintenance

In Chapter 8 it was shown: small steps counteract entropy. Here it is described what entropy is in the context of hierarchical systems, where it comes from, and how to work with it.

9.1 Entropy as a System Property

Entropy is the tendency of a system to degrade without an inflow of energy. Not to collapse. To degrade. The system does not fall apart instantly. It gradually loses quality: information is forgotten, connections cool, people leave, newcomers do not know the history, processes lose meaning.

This is not a defect. This is a property of any system that exists in time. Without an inflow of energy, any structure decays. Physical, biological, social — any.

In the context of this model, entropy means: if no maintenance actions are taken, every dimension of every axis will gradually decline. Not in a jump. Not immediately. But inevitably.

9.2 Two Sources of Entropy

Entropy comes from two sources.

Internal entropy. Processes within the system that lead to degradation without external impact.

People forget. A contact that was substantive six months ago becomes formal if it is not maintained. A language that was shared diverges if teams stop interacting. Trust that was deep cools if there are no joint actions.

People leave. A key contact moves to another position. A carrier of the shared language leaves the organization. Agency that was distributed concentrates in one person, and upon their departure the system loses a level.

Processes lose meaning. A procedure that was alive becomes a ritual. A meeting that was a platform for decisions becomes a report. A project that was joint becomes a formality.

External entropy. Changes in the environment that destroy the system from outside.

Competition. A new participant appears that offers the same thing but faster, cheaper, or more conveniently. If the system does not adapt, its connections lose depth.

Regulatory changes. Rules, standards, requirements change. If the system does not embed new requirements, its quality ceases to correspond to the environment.

Technological shifts. A new technology appears that changes the mode of interaction. If the system does not adopt it, its topology becomes obsolete.

Change of key figures. A leader, owner, or curator changes. The new person does not have the history. If the system has not embedded knowledge in structure rather than in people, it loses a level with each change.

9.3 Maintenance as a Continuous Process

Maintenance is not a one-time action. It is a continuous process. One cannot “maintain once” and forget. Entropy does not stop. Maintenance must not stop either.

In Chapter 8 the principle of proportional support was described: the higher the level, the more substantial the small steps. Here a second rule is added: maintenance must be regular, not episodic.

Regularity does not mean frequency. At level 1, one touch per quarter is sufficient. At level 4 — weekly interaction. But in both cases this is regularity, not randomness. A system that receives support once a year degrades between touches. A system that receives support on a rhythm maintains the level.

9.4 Antifragility

Not all systems react to stress equally. A fragile system is destroyed by stress. A robust system withstands stress but does not change. An antifragile system is strengthened by stress.

In the context of this model, antifragility means: a stressful event can be not destructive but transitional. If the system is ready — if potential is accumulated, synchronization exists, levels are high — stress triggers a transition to the next level. If the system is not ready — stress triggers a cascading collapse.

The difference is not in the stress. The difference is in the state of the system. The same crisis for one system is a catastrophe, for another — an opportunity. Not because the crisis is different. Because the systems are different.

Example. The 2020 pandemic. For a city with a diversified economy, distributed agency, and deep connections between groups — stress that accelerated the transition. For a city dependent on one factory, with formal connections and concentrated agency — stress that triggered a cascading collapse. The crisis is the same. The result is different. The difference is in the state of the system before the crisis.

Example. Change of leadership. For an organization with a shared language, embedded processes, and distributed agency — stress that does not change the level. For an organization where everything depended on one person — stress that collapses the system. The event is the same. The result is different. The difference is in the structure.

9.5 Connection with Phases

Entropy pulls the system down. Small steps counteract. Events transition upward.

These are three forces that act simultaneously. The system at any moment is under the influence of all three. If small steps are sufficient — the system maintains the level. If events occur — the system grows. If neither exists — the system degrades.

A phase transition upward requires an event. A phase transition downward can occur without an event — simply from cessation of maintenance. This is an asymmetry: growth requires effort, degradation happens by itself.

9.6 Summary Tables

Force	Direction	Condition	Result
Entropy	Downward	Absence of maintenance	Degradation
Small steps	Maintenance	Regularity, proportionality	System remains at level
Events	Upward	Independent value, shift in perception	System transitions to next level
Stress in a ready system	Upward	Potential accumulated, synchronization exists	Antifragility: stress transitions
Stress in an unready system	Downward	Potential not accumulated, desynchronization	Fragility: stress destroys

Source of entropy	Example	Action
Internal: forgetting	Contact cools without maintenance	Regular small steps
Internal: people leave	Carrier of language leaves the system	Embedding knowledge in structure, not in people
Internal: loss of meaning	Procedure becomes a ritual	Periodic review of processes
External: competition	New participant offers better	Adaptation, innovation
External: regulatory changes	New requirements	Embedding in processes
External: change of key figures	New leader without history	Documentation, transfer of context

Summary of Part II. Rules

Part II describes five rules that operate in the space defined by Part I. The rules are not recommendations. They are consequences of structure.

Rule	Chapter	Essence
Synchronization	5	Spread within axis ≤ 1 . Spread between axes ≤ 1 . Priority to the lagging dimension.
Field separation	6	Z, Y, X — action field. W — result field. Do not manage W directly. W will follow.
Structural ceiling	7	Objective constraint of the environment. Three cycles without shift — ceiling. Strategies: alignment, expansion.
Small steps and events	8	Small steps maintain. Events transition. Catalyst — a small step in a pre-critical system.
Entropy and maintenance	9	Without maintenance the system degrades. Maintenance is regular and proportional. Stress transitions a ready system.

These rules operate simultaneously. Synchronization shows where desynchronization exists. Field separation shows where to direct effort. Structural ceiling shows where effort will not yield results. Small steps and events show how to act. Entropy shows what happens without action.

PART III

DYNAMICS

10 Chapter 10. Phase Transitions

In Part I the space was described: axes, dimensions, aspects, phases. In Part II the rules were described: synchronization, field separation, structural ceiling, small steps and events, entropy. Here it is described how the system moves in this space. Not maintains. Moves.

A phase transition is a jump of the system from one phase to another. Not a smooth movement. A jump. The system was in phase 1. Became in phase 2. There was no intermediate state.

10.1 Three Levels of Transition

A phase transition can occur at three levels.

Level one: transition within a dimension.

One dimension changes phase. For example, X_2 (depth of trust) transitions from level 1 (formal trust) to level 2 (expert trust). The other dimensions remain at their previous levels.

This is a minimal transition. It changes the state of one element of the system but does not change the system as a whole. Axis X can remain at an average level of 1.67 even if X_2 has transitioned to 2.

Level two: transition of an axis.

All three dimensions of an axis transition to the next level. For example, axis X: X_1 , X_2 , X_3 — all transition from level 1 to level 2. The axis as a whole changes phase.

This is a transition that changes the quality of the axis. The network becomes not just broad but alive. Trust becomes not just formal but substantive. The flow becomes not just regular but dense. The axis transitions to a new phase as a single whole.

Level three: coupling.

Several axes transition simultaneously or in rapid succession. For example, Z and X transition from level 2 to level 3 within one cycle. The system changes not one axis but the configuration as a whole.

This is a transition that changes the system. Not a separate element, not a separate axis, but the entire configuration. The system was in one state. Became in another. Not gradually. In a jump.

10.2 Mechanics of Transition

Transition at each level has its own mechanics.

Transition within a dimension is triggered by an event that changes one specific quality. Not the entire axis. One dimension. For example, joint resolution of a specific task can transition X_2 (depth of trust) from level 1 to level 2 without changing X_1 (breadth) and X_3 (density).

Transition of an axis requires synchronization of all three dimensions. If one dimension lags, the axis does not transition. This is a direct consequence of the

weak link rule: the lagging element does not allow rising higher. Transition of an axis occurs when all three dimensions reach the critical point simultaneously or in rapid succession.

Coupling requires synchronization between axes. If one axis transitions while others lag, coupling does not occur. The system receives desynchronization. Coupling occurs when several axes reach the critical point in one time window.

10.3 Synchronization During Transition

During a phase transition, lagging dimensions are pulled up. Not always to the same level. But into the zone of spread ≤ 1 .

This is not an automatic process. Pulling up requires effort. If axis Z transitions to level 3 while axis X remains at level 1, the system is not stable. Spread = 2. Desynchronization. Priority — X. Not Z. Not Y. X.

Pulling up can occur in two ways. First: the lagging axis grows independently, through its own events. Second: the growing axis pulls the lagging one along, if there is a connection between them. For example, growth of Z (language, vision) can pull X (connections), because a shared language creates new channels. But this is not guaranteed. The connection between axes exists, but it is not automatic.

10.4 W During Transition

W during a phase transition behaves differently from Z, Y, X. W does not transition simultaneously with them. W lags.

This is a direct consequence of field separation described in Chapter 6. Z, Y, X — action field. W — result field. During transition, Z, Y, X change first. W changes last. With a delay.

The delay of W during transition is not a problem. This is the norm. If Z, Y, X have transitioned to level 3 while W remains at level 1 — this is a growth zone. Potential is accumulated. W will follow. The question is only one of delay.

But if W does not follow within several cycles after the transition of Z, Y, X — this is a signal. Either the transition of Z, Y, X was assessed incorrectly. Or there exists a structural ceiling on W. Or an external impact has occurred that blocks the realization of potential.

10.5 The Z → Y/X → W Pattern

Phase transitions in hierarchical systems follow a stable pattern:

Z transitions first. Identity changes before everything else. Language, vision, agency change. The system begins to understand itself differently. This happens before quality, connections, and resources change.

Y and X transition next. Quality and connections change after identity. When the system understands where it is going, it begins to create different quality and build different connections. Y and X can transition simultaneously or in rapid succession.

W transitions last. Resources change after everything. When identity, quality, and connections are already in a new phase, resources follow. With a delay. Not simultaneously.

This pattern is not a recommendation. This is an observed regularity. It reproduces in different domains: in organizational transformations, in urban developments, in partnerships, in products. Not because someone decided so. But because identity defines the landscape, quality and connections fill the landscape, resources are projected onto the filled landscape.

Violation of the pattern — an attempt to start with W — leads to the risk zone described in Chapter 6. Resources without identity, quality, and connections are exchange without structure. Exchange without structure does not create connection. It creates dependency.

10.6 Examples from Different Domains

Domain: healthcare. Formation of an interdisciplinary treatment model.

Z transitions first: a medical institution formulates a new treatment model, creates a shared language between specialists of different profiles, defines the agenda. Y and X transition next: new protocols are embedded in everyday practice, interdisciplinary teams form, trust between doctors and patients deepens. W transitions last: funding arrives, patient flow grows, indicators improve.

An attempt to start with W — obtaining funding before forming a shared model — leads to formal reporting without changing practice. Resources exist, but identity does not. Risk zone.

Domain: sport / team. Formation of a playing system.

Z transitions first: a team formulates a playing philosophy, creates a shared language, defines roles and agency of each player. Y and X transition next: tactics are embedded in the game, connections between players form, trust on the field deepens. W transitions last: victories come, audience grows, revenues increase.

An attempt to start with W — buying expensive players before forming a philosophy — leads to a collection of individuals without a system. Resources exist, but identity does not. Risk zone.

Domain: urban management. Development of a city.

Z transitions first: a city formulates a vision of the future, creates a shared language, distributes agency. Y and X transition next: projects are embedded in the environment, connections between groups form, trust deepens. W transitions last: the economy diversifies, external resources arrive, resilience grows.

An attempt to start with W — attracting an investor before forming identity — leads to dependence on a single source. Resources exist, but the city does not understand where it is going. Risk zone.

10.7 Summary Tables

Level of transition	What changes	Condition	Result
Dimension	One dimension	Event that changes one quality	Minimal transition
Axis	All three dimensions of an axis	Synchronization of dimensions	Axis changes phase
Coupling	Several axes	Synchronization between axes	System changes configuration

Pattern	Order	Property
$Z \rightarrow Y/X \rightarrow W$	Identity first. Resources last.	Observed regularity, not a recommendation
W without Z, Y, X	Resources without structure	Risk zone. Exchange without connection.
Z, Y, X without W	Potential without realization	Growth zone. W will follow.

11 Chapter 11. Reverse Phase Transition

In Chapter 10 it was described how the system transitions upward: from phase to phase, from dimension to axis, from axis to coupling. Here the reverse movement is described. Not growth. Collapse.

A reverse phase transition is a jump of the system from a higher phase to a lower one. Not a smooth degradation. A jump. The system was in phase 3. Became in phase 1. There was no intermediate state.

11.1 Mechanics of Cascade

A reverse transition rarely affects one dimension. The fall of one dimension pulls others. This is a cascade.

The mechanics of cascade follow from structure. Dimensions within an axis are connected. Axes between each other are connected. If one dimension falls, it creates load on adjacent ones. If adjacent ones do not withstand — they fall too.

Example. Axis X: $X_1 = 3$, $X_2 = 3$, $X_3 = 3$. The network is broad, trust is deep, the flow is active. A key contact leaves the organization.

At level 3, trust is not tied to one person. It is embedded in structure. The departure of one contact does not collapse X_2 instantly. The system is stable. But stability is not a guarantee. It is time.

If the system does not adapt — does not redistribute roles, does not embed the knowledge of the departed into structure, does not create new channels — entropy does its work. X_2 degrades from 3 to 2 over several cycles. Trust ceases to be self-reproducing. It becomes dependent on the remaining people.

Then cascade. X_2 at level 2 does not hold X_3 : without deep trust, the flow declines. X_3 falls from 3 to 1. X_1 still holds: the network formally exists. But without trust and flow, the network becomes formal. X_1 falls from 3 to 1. Axis X has collapsed entirely.

The cascade does not stop at the axis. The fall of X pulls Y: without connections, quality is not embedded. Y falls. The fall of Y pulls Z: without quality, vision loses meaning. Z falls. The system collapses not because all axes were weak. But because one weak link did not withstand the load of time.

This is a direct consequence of the weak link rule formulated in Chapter 1. During growth, the weak link does not allow rising. During collapse, the weak link pulls down.

Level 3 protects against instant collapse. But it does not protect against degradation without maintenance. The cascade is slowed but not canceled. The difference between level 2 and level 3 during a reverse transition is not the difference between “will collapse” and “will not collapse.” It is the difference between “will collapse immediately” and “will collapse over several cycles if not adapted.”

11.2 W as a Lagging Indicator of Danger

W during a reverse transition behaves the same as during a direct one: with a delay. But in the reverse direction this creates a special danger.

When Z, Y, X begin to fall, W still holds. Resources continue to flow. Contracts are active. Budgets are approved. From the outside, everything seems fine. W shows well-being.

But Z, Y, X are already falling. Language diverges. Quality degrades. Connections cool. W has not yet reacted because resource decisions are inert. The contract is signed for a year. The budget is approved for a cycle. The partner does not know that trust is already lost.

When W finally falls — it is already too late. Z, Y, X by this moment are at levels from which recovery requires not small steps but events. Sometimes — events that are impossible in the current configuration.

This is an asymmetry of danger. During a direct transition, the delay of W is a growth zone: potential is accumulated, resources will follow. During a reverse transition, the delay of W is a zone of false well-being: resources still exist, but structure is already destroyed.

11.3 The Monitoring Rule

From the lag of W follows a rule:

Monitoring is conducted on Z, Y, X. Not on W.

W shows what has already happened. Z, Y, X show what is happening. If monitoring is conducted only on W, the system learns about the problem when it is already too late to solve it.

This does not mean that W should not be tracked. W is needed. But W is confirmation, not signal. The signal is in Z, Y, X. Confirmation is in W.

Practically: if Z, Y, X are falling while W holds — this is not well-being. This is a reprieve. Actions are needed immediately, not when W falls. If Z, Y, X are growing while W holds — this is not stagnation. This is a delay. Actions are not needed, patience is.

11.4 Speed of Reverse Transition

A reverse transition is faster than a direct one. This is not an accident. This is a property of structure.

A direct transition requires an event: accumulation of potential, critical point, jump. This is slow. Potential accumulates over cycles. The event occurs in a moment. The transition is recorded.

A reverse transition can occur without an event. Cessation of maintenance is sufficient. Entropy, described in Chapter 9, does its work. The system degrades by itself. No destructive impact is needed. One simply needs to stop maintaining.

This is an asymmetry: growth requires effort, degradation happens by itself. A direct transition is work. A reverse transition is the absence of work.

11.5 Examples from Different Domains

Domain: M&A. Collapse after failed integration.

The deal is closed. W is high: budgets are consolidated, resources are merged. But Z, Y, X were not built: teams do not know each other, language is different, trust is formal. W holds on the inertia of the contract.

After two cycles, a key manager of one of the teams leaves. X₂ falls: trust was tied to him. Cascade: X₃ falls (flow stops), Y falls (joint projects stop), Z falls (shared language diverges). W falls last: after six months, when the contract is revised. By this moment, recovery is impossible. Teams are already in the “us — them” model.

Domain: product / user. Loss of user base.

The product is on the market. W is high: revenue grows, market share increases. But Z, Y, X are not maintained: vision is not updated, quality is not adapted to new needs, feedback channels are formal. W holds on the inertia of habit.

A competitor appears with a more adaptive product. X₂ falls: users begin to try the alternative. Cascade: X₃ falls (frequency of use declines), Y falls (the product ceases to be embedded in everyday life), Z falls (users cease to identify themselves with the product). W falls last: after several cycles, when subscriptions are not renewed. By this moment, returning users is more expensive than acquiring new ones.

Domain: education / research school. Collapse of a research direction.

The school is recognized. W is high: grants, citations, student flow. But Z, Y, X are not maintained: the paradigm is not updated, methods are not adapted, collaborations are formal. W holds on reputation.

The leader of the school retires. Z₂ falls: agency was concentrated in one person. Cascade: Z₃ falls (vision is not shared), X falls (collaborations fall apart), Y falls (methods become obsolete). W falls last: after several years, when grants are not renewed. By this moment, the school no longer exists as a single whole. Individual participants work in other paradigms.

11.6 Summary Tables

Parameter	Direct transition	Reverse transition
Requires	Event	Cessation of maintenance
Speed	Slow (accumulation of potential)	Fast (entropy)
W	Lags (growth zone)	Lags (zone of false well-being)
Monitoring	On Z, Y, X	On Z, Y, X
Recovery	Possible with small steps and events	Requires events, sometimes impossible

Stage of cascade	What happens	Signal
Fall of one dimension	Weak link did not withstand	Spread within axis ≥ 2
Fall of an axis	Cascade within axis	Axis mean falls by 1 +
Fall of several axes	Inter-axis cascade	Spread between axes ≥ 2
Fall of W	Confirmation of destruction	W falls. Already too late.

12 Chapter 12. Trajectories and Cycles

In Chapters 10 and 11 it was described how the system transitions upward and downward. Here it is described how to track the movement of the system over time. Not in a moment. In dynamics.

12.1 Trajectory

A trajectory is a sequence of snapshots of a system over time. One snapshot shows state. A trajectory shows movement.

A snapshot records twelve numbers: four axes, three dimensions, levels from 0 to 4. A trajectory records a sequence of such snapshots. Not two. Not three. At least four to five to see direction.

A trajectory is not a forecast. It does not say what will happen. It says where the system is moving. The difference is fundamental. A forecast requires knowledge of future events. A trajectory requires only knowledge of past states.

12.2 Cycle

A cycle is the period between snapshots. Not arbitrary. Tied to the natural rhythm of the specific system.

A cycle is determined not by the type of system but by the speed of processes within it. The same system can require different cycles at different stages. A city in a phase of active transformation may require a snapshot once a year. The same city in a phase of plateau — once every three years. A simple digital product may require a snapshot once a month. A spacecraft — once every several years.

There is no universal cycle. There is an optimal cycle for a specific system in a specific state.

A cycle that is too short gives noise: changes have not yet occurred, but the snapshot already records. A cycle that is too long gives blindness: changes have occurred, but the snapshot does not catch them. The optimal cycle is one during which the system has time to respond to interventions but does not have time to degrade without them.

12.3 How to Read a Trajectory

A trajectory is read by three parameters: speed, direction, desynchronization.

Speed. How quickly levels change. If over three cycles an axis grew by 1 — this is slow growth. If over one cycle — fast. Speed is not an evaluation. Slow growth is not bad. Fast growth is not good. Speed shows how much time the system has for adaptation.

Direction. Where the system is moving. Up, down, sideways. Direction is determined not by one snapshot but by a sequence. If over three cycles the mean on Z grew from 1.33 to 2.00 to 2.67 — direction is up. If from 2.67 to 2.00 to 1.33 — direction is down. If 2.00, 2.00, 2.00 — there is no direction. Stagnation.

Desynchronization. How the spread between axes and within axes changes. If the spread grows — the system loses coherence. If the spread falls — the system synchronizes. Desynchronization on a trajectory is visible earlier than in a single snapshot. One snapshot shows the spread. A trajectory shows whether it grows or falls.

12.4 Typical Trajectories

Four typical trajectories. Not an exhaustive list. But covering the majority of observed cases.

Growth trajectory. Levels grow. Spread is within norm. W lags but follows. The system moves upward in concert.

Indicator: each cycle the mean on Z, Y, X grows by 0.3–0.5. Spread between axes ≤ 1 . W grows with a delay of 1–2 cycles.

Action: maintenance. Do not force. Do not change strategy. The system works.

Stagnation trajectory. Levels do not change. Spread is within norm. W does not change. The system stands still.

Indicator: over three cycles the mean on Z, Y, X has not changed. Spread ≤ 1 . W does not change.

Action: diagnostics. Stagnation can be a structural ceiling (Chapter 7) or a lack of effort. If effort is conducted methodically and levels do not change for three cycles — likely a ceiling. If effort is not conducted — stagnation is cured by resuming work.

Cascade trajectory. Levels fall. Spread grows. W holds, then falls. The system is being destroyed.

Indicator: over one to two cycles one or several dimensions fall by 1+. Spread within axis ≥ 2 . W still holds. Then W falls.

Action: urgent. Priority to the lagging dimension. Do not wait for W to fall. If W has fallen — recovery requires events, sometimes impossible ones.

Plateau trajectory. Levels have grown and stabilized. Spread is within norm. W is stable. The system has reached a ceiling and holds it.

Indicator: over three cycles the mean does not change. Spread ≤ 1 . W is stable. Effort is conducted but there is no growth.

Action: alignment or expansion (Chapter 7). If all axes are at the ceiling — expansion into an adjacent domain. If one axis is at the ceiling while others lag — alignment.

12.5 Examples from Different Domains

Domain: organizational management. Internal transformation of a company.

Growth trajectory: over four quarters Z grew from 1.00 to 2.33, Y from 1.00 to 2.00, X from 1.33 to 2.33. W grew from 0.33 to 1.00 with a delay of two quarters. Spread ≤ 1 . The system moves upward in concert. Action: maintenance.

Cascade trajectory: over one quarter a key leader leaves. Z_2 falls from 3 to 1. Spread within $Z = 2$. W still holds. Action: urgent. Redistribution of agency, embedding knowledge in structure. Do not wait for W to fall.

Domain: military structure. Formation of a combat-capable unit.

Growth trajectory: over three cycles of exercises Z grew from 1.00 to 2.00 (shared language, understanding of the task), X from 1.00 to 2.33 (joint actions, trust). Y grew from 1.33 to 2.00 (tactics are embedded). W is stable (supply by protocol). Spread ≤ 1 . Action: continuation of exercises.

Plateau trajectory: the unit has reached level 3 on Z , Y , X . Further exercises do not yield growth. Ceiling on Y : tactics do not become more complex without new armament. Action: expansion — mastering adjacent tasks, interaction with other units.

Domain: creative collective. Formation of an artistic school.

Stagnation trajectory: over two years Z has not changed (language is formed but does not develop), Y has not changed (methods are the same), X has not changed (composition is the same). W is stable (exhibitions, sales). Effort is conducted but there is no growth. Action: diagnostics. Likely a structural ceiling on Y : methods are exhausted. Expansion: entry into adjacent media, collaborations with other schools.

Cascade trajectory: the leader of the collective changes direction. Z_3 falls (vision is not shared). Cascade: X falls (participants disperse), Y falls (methods lose meaning). W falls last (exhibitions cease). Action: if Z_3 is not restored — the collective falls apart. Recovery requires an event: new vision, new leader, new paradigm.

12.6 Summary Tables

Trajectory	Indicator	Spread	W	Action
Growth	Levels grow each cycle	≤ 1	Lags, follows	Maintenance
Stagnation	Levels do not change 3+ cycles	≤ 1	Does not change	Diagnostics: ceiling or lack of effort
Cascade	Levels fall, spread grows	≥ 2	Holds, then falls	Urgent: priority to lagging
Plateau	Levels stabilized	≤ 1	Stable	Alignment or expansion

Reading parameter	What it shows	How to record
Speed	How quickly levels change	Δ level per cycle
Direction	Where the system is moving	Sequence of means

Desynchronization	How coherence changes	Dynamics of spread
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13 Chapter 13. Strategies in the Absence of Growth

In Chapters 10–12 it was described how the system grows, falls, and moves over time. Here a situation is described in which growth is impossible and maintenance is not expedient. Not stagnation. Not a ceiling. Exhaustion.

When a system has reached a structural ceiling, efforts toward growth yield no result. When entropy exceeds maintenance capacity, the system degrades despite effort. When the environment has changed irreversibly, the previous configuration cannot exist in the new environment.

In these cases, three strategies are possible: freeze, managed downgrade, closure.

13.1 Freeze

A freeze is the fixation of the system at the current level without attempts at growth. Maintenance continues but is minimal. The goal is not to grow. The goal is not to degrade.

A freeze is applied when the system has reached a ceiling but the current level is valuable. Growth is impossible, but degradation is undesirable. The system is needed in its current state.

During a freeze, efforts toward growth cease. Events are not planned. Small steps continue but in minimal volume — only to counteract entropy. Monitoring is conducted on Z, Y, X. If Z, Y, X begin to fall — the freeze is not working. Revision is required.

A freeze is a strategy of retention. A system at level 3 that cannot grow to 4 but stably holds 3 is a stable system. Not growing. But stable.

Example. A division of a company has reached level 3 on all axes. Further growth is impossible: the role does not presuppose strategic partnership. Freeze: maintenance of the current level, minimal small steps, monitoring. The division is stable. Does not grow. But does not degrade either.

13.2 Managed Downgrade

A managed downgrade is a deliberate transition of the system to a lower level. Not a cascade. Not a collapse. A managed process.

A managed downgrade is applied when the current level cannot be maintained. Resources are shrinking. The environment has changed. Maintaining the current level requires effort that exceeds what is available. Lowering the level reduces maintenance requirements.

During a managed downgrade, the system is deliberately transitioned to a level that can be maintained with available resources. Not to zero. To a level that is stable.

Example. A city depended on one factory. The factory reduces production. W falls. Maintaining the current level of Z, Y, X is impossible: resources do not allow it. Managed downgrade: the city deliberately transitions to level 2 on Z, Y, X.

Language is simplified. Projects are reduced. Connections narrow. But the system remains stable at level 2. Not at level 3, which it cannot maintain. And not at level 0, to which a cascade would lead.

A managed downgrade is adaptation. The system chooses a level it can hold instead of a level it cannot. The difference between a managed downgrade and a cascade is the presence of a decision. A cascade happens by itself. A managed downgrade happens by decision.

13.3 Closure

Closure is the cessation of the system's existence. Not a downgrade. Not a freeze. Completion.

Closure is applied when the system is exhausted. The environment has changed irreversibly. The configuration cannot exist in new conditions. Maintenance is not expedient. A downgrade does not yield stability. The system is not needed.

During closure, the system is deliberately completed. Resources are redistributed. Knowledge is documented. Connections are transferred. Closure is not destruction. It is completion with preservation of what can be used in other systems.

Example. A research school has exhausted its paradigm. New directions do not emerge. Young researchers leave for other fields. Maintaining the school requires resources that yield no result. Closure: the school is deliberately completed. The archive is transferred to a university. Methods are documented. Participants move to other schools. The school ceases to exist as a single whole. But its knowledge is preserved.

Closure is the recognition that the system has fulfilled its function. Not every system must exist forever. Some systems exist to create conditions for other systems.

13.4 Criteria of Choice

The three strategies are chosen by two criteria: possibility of maintenance and value of the current state.

Criterion	Freeze	Managed downgrade
Closure		

	Freeze	Managed downgrade	Closure
Current level is valuable	Yes	Possibly	No
Maintenance is possible	Yes	No	No
Growth is possible	No	No	No
System is needed	Yes	At lower level	No

Freeze is chosen when the current level is valuable and can be maintained. Growth is impossible, but degradation is undesirable. The system is needed as is.

Managed downgrade is chosen in two cases. First: the current level is valuable but cannot be maintained. The system is lowered to a level it can hold. Second: the current level is not valuable, but the system can be useful at a lower level.

Closure is chosen when the current level is not valuable and cannot be maintained. The system is exhausted. Continued existence yields no value.

13.5 Examples from Different Domains

Domain: healthcare. Interdisciplinary treatment program.

Freeze: the program has reached level 3. Further growth is impossible: regulatory constraints do not allow expanding the protocol. Freeze: maintenance of the current level, minimal updates, monitoring. The program is stable. Does not grow. But works.

Managed downgrade: funding is shrinking. Maintaining level 3 is impossible. The program deliberately transitions to level 2: interdisciplinary teams are reduced, protocols are simplified. The system remains stable at level 2. Not at level 3, which it cannot maintain.

Closure: the program is exhausted. The methodology is obsolete. New approaches make it irrelevant. Closure: the program is completed. Data is transferred to a research center. Participants move to new programs.

Domain: sport / team. Playing system.

Freeze: the team has reached level 3. Further growth is impossible: the roster does not allow complicating tactics. Freeze: maintenance of the current level, minimal training, monitoring. The team is stable. Does not grow. But is competitive.

Managed downgrade: key players leave. Maintaining level 3 is impossible. The team deliberately transitions to level 2: tactics are simplified, roles are redistributed. The system remains stable at level 2. Not at level 3, which it cannot maintain.

Closure: the team is exhausted. Players have left. The coach has changed. Funding has ceased. Closure: the team is disbanded. Infrastructure is transferred to another team. History is documented.

Domain: urban management. City development program.

Freeze: the program has reached level 3. Further growth is impossible: the scale of the city does not allow attracting federal resources. Freeze: maintenance of the current level, minimal projects, monitoring. The program is stable. Does not grow. But works.

Managed downgrade: the factory reduces funding. Maintaining level 3 is impossible. The program deliberately transitions to level 2: projects are reduced, connections narrow. The system remains stable at level 2. Not at level 3, which it cannot maintain.

Closure: the program is exhausted. The city has changed. The previous model is not relevant. Closure: the program is completed. Knowledge is transferred to the administration. Participants move to new initiatives.

13.6 Summary Tables

Strategy	Condition	Action	Result
Freeze	Level is valuable, maintenance is possible	Minimal maintenance, monitoring	System is stable at current level
Managed downgrade	Level cannot be maintained or is not valuable	Deliberate transition to lower level	System is stable at available level
Closure	System is exhausted, continuation yields no value	Deliberate completion	System is completed, knowledge is preserved

Summary of Part III. Dynamics

Part III describes how the system moves in the space defined by Part I, by the rules described in Part II.

Chapter	Topic	Essence
10	Phase transitions	Three levels: dimension $\rightarrow$ axis $\rightarrow$ coupling. Pattern $Z \rightarrow Y/X \rightarrow W$. Identity first. Resources last.
11	Reverse phase transition	Cascade: fall of one dimension pulls others. W lags — zone of false well-being. Monitoring on Z, Y, X.
12	Trajectories and cycles	Trajectory is a sequence of snapshots. Cycle is determined by speed of processes. Four typical trajectories: growth, stagnation, cascade, plateau.
13	Strategies in the absence of growth	Freeze, managed downgrade, closure. Choice by two criteria: possibility of maintenance, value of current state.

These chapters describe movement. Chapter 10 — upward. Chapter 11 — downward. Chapter 12 — over time. Chapter 13 — when movement stops.

PART IV

DIAGNOSTICS

14 Chapter 14. Snapshot

In Part I the space was described: axes, dimensions, aspects, phases. In Part II the rules were described. In Part III the dynamics were described. Here the instrument is described that turns the model into practice. The snapshot.

14.1 Definition

A snapshot is a one-time recording of the state of a system across all twelve dimensions. Not an evaluation. Not an opinion. A recording.

A snapshot answers the question: where is the system now? Not where it is going. Not where it should go. Where it is now.

One snapshot is twelve numbers. Four axes. Three dimensions in each. A level from 0 to 4 for each dimension. Total: $Z_1, Z_2, Z_3, Y_1, Y_2, Y_3, X_1, X_2, X_3, W_1, W_2, W_3$.

A snapshot is not a diagnosis. A snapshot is data. A diagnosis is built on the basis of data: spread, desynchronization, direction. But without a snapshot there is no diagnosis. A snapshot is the primary instrument of observation.

14.2 What Is Recorded

For each dimension, a level from 0 to 4 is recorded. The level is determined not by opinion but by an observable event.

Level 0: an event corresponding to this quality has not been recorded. The quality is absent.

Level 1: an event corresponding to formal structure has been recorded. Form exists, movement does not.

Level 2: an event corresponding to development has been recorded. Form adapts, exchange is substantive.

Level 3: an event corresponding to stable state has been recorded. Self-reproduction without external impulse.

Level 4: an event corresponding to self-reproduction has been recorded. The system generates quality independently.

For each level there exists a set of observable criteria. Criteria differ for different domains, but the structure is uniform: an event occurred or did not occur. If it occurred — the level is confirmed. If it did not occur — the level is not confirmed.

14.3 How It Is Recorded

A snapshot is recorded through observable events. Not through opinions. Not through feelings. Not through self-assessment.

An observable event is a fact that can be recorded by an external observer. A meeting took place. A document was signed. A project was launched. A decision was made jointly. A conflict was resolved without escalation. A new participant came by recommendation.

An opinion is a judgment that cannot be verified. “I feel that trust has grown.” “I sense that the team has become more cohesive.” “I think we are on the right track.” Opinions are not recorded. Opinions are not data.

The difference is fundamental. An opinion admits an infinite range of interpretations. An event admits two answers: it occurred or it did not. A snapshot built on opinions is not reproducible. A snapshot built on events is reproducible. Two observers recording the same events will obtain the same snapshot. Two observers recording the same opinions will obtain different snapshots.

14.4 Presumption of Unconfirmed

If an observable event corresponding to the criteria of a level has not been recorded — the system remains at the current confirmed level. Not at the assumed one. At the confirmed one.

This means: the burden of proof lies with the one who asserts a higher level. If there is no event confirming level 3, the system is at level 2. Not “between 2 and 3.” At 2.

The presumption of unconfirmed removes subjectivity. It does not allow inflating levels based on expectations, hopes, or interpretations. A level is confirmed by an event. No event — no level.

14.5 Example of a Completed Snapshot

Domain: M&A. Coupling of two organizations after a deal. Snapshot six months after closing.

Dimension	Level	Confirming event
Z ₁ . Language	2	Teams use shared terminology in working documents. Recorded in three joint reports.
Z ₂ . Agency	1	Decisions are made by the leadership of the combined structure. Teams execute, do not initiate.
Z ₃ . Vision	2	A joint strategic session was held. Vision is formulated but not shared by all.
Y ₁ . Embeddedness	1	Processes are integrated formally. Protocol is signed. Everyday practice has not changed.
Y ₂ . Adaptation	1	Proposals of one team do not account for the specifics of the other. Standard solutions without customization.
Y ₃ . Innovation	0	No joint developments. No new formats have emerged.
X ₁ . Breadth	2	Three joint working groups have been created. Channels exist.
X ₂ . Depth	1	Interaction by protocol. Outside working groups, contacts are minimal.

X ₃ . Density	2	Meetings are weekly. Document exchange is regular.
W ₁ . Share	1	Budgets are consolidated, but distribution is through one center.
W ₂ . Resilience	1	Upon departure of a key manager of one of the teams, the process will stop.
W ₃ . Duration	2	Integration contract is signed for two years.

Means: $Z = 1.67$, $Y = 0.67$, $X = 1.67$, $W = 1.33$.

Spread within axes: $Z = 1$, $Y = 1$, $X = 1$, $W = 1$. All axes are synchronized.

Spread between axes: $\max(Z, X) = 1.67$, $\min(Y) = 0.67$. Spread = 1.00. The system is synchronized.

Diagnosis: the system is at an early stage of integration. Z and X lead Y. Y lags: quality is not embedded, adaptation is formal, there are no innovations. Priority: Y. Not Z, not X. Y.

14.6 What a Snapshot Shows and What It Does Not

A snapshot shows state. Not cause. Not forecast. Not recommendation.

A snapshot shows: levels across twelve dimensions. Spread within axes. Spread between axes. Zone: growth, risk, synchronization.

A snapshot does not show: why the system is in this state. What led to the current levels. What events were. What people participated. What the history is.

A snapshot does not show: where the system will go. Whether there will be growth. Whether there will be a cascade. When W will change.

A snapshot does not show: what to do. What actions to take. What events to plan. Whom to involve.

For this, other instruments are needed: trajectory (Chapter 12), diagnostic atlas (Chapter 15), journal (Chapter 16). A snapshot is the first step. Without it, the other instruments do not work. But a snapshot by itself is not a solution.

14.7 Summary Table

Parameter	Description
What is recorded	12 dimensions, levels 0-4
How it is recorded	Observable events, not opinions
Presumption	Unconfirmed: no event — no level
What it shows	State, spread, zone
What it does not show	Cause, forecast, recommendation
Reproducibility	Two observers, same events — one snapshot

15 Chapter 15. Diagnostic Atlas

In Chapter 14 it was described how a snapshot is recorded: twelve dimensions, levels from 0 to 4, observable events. Here the instrument for determining the level is given. The diagnostic atlas.

15.1 Definition

The diagnostic atlas is a table of observable criteria for each dimension at each level. 12 dimensions $\times$ 5 levels = 60 cells. Each cell contains a description of what is observed if the system is at this level on this dimension.

The atlas is not a test. Not a questionnaire. Not a self-assessment checklist. The atlas is a reference for an observer. The observer records events. Then opens the atlas. Finds the dimension. Finds the level. Checks: does the recorded event correspond to the criteria of the level? If yes — the level is confirmed. If no — the level is not confirmed.

15.2 How to Use

Three steps.

Step one: find the dimension. Determine which of the twelve dimensions is being assessed. Z₁, Z₂, Z₃, Y₁, Y₂, Y₃, X₁, X₂, X₃, W₁, W₂, W₃. Each dimension has a name and an aspect. The name indicates what is assessed. The aspect indicates what question is asked: how it is structured (topology), what quality it possesses (state), how it moves (dynamics).

Step two: find the level. Start from level 0 and move upward. For each level, check: has an event corresponding to the criteria been recorded? If no — the system is at the previous level. If yes — move to the next. Stop at the first level whose criteria are not confirmed. The previous level is the current one.

Step three: record. Write down the level and the confirming event. Not the interpretation. The event. Not “trust has grown.” But “the buyer consulted the supplier on specification.” Not “the team has become cohesive.” But “the division independently proposed a solution that was accepted without corrections.”

15.3 Principle

Do not interpret. Record.

Interpretation is a judgment about the meaning of an event. Recording is the writing down of the event itself. The atlas works with recording. Meaning is assigned later, when building a diagnosis. At the stage of determining the level, meaning is not needed. Only this is needed: did the event occur or did it not.

This removes subjectivity. Two observers recording the same events will obtain the same level. Two observers interpreting the same events will obtain different levels. The atlas is built on the first, not the second.

15.4 Axis Z — Identity

How the system understands itself, its past, and its future.

15.4.1 Z₁. Language (topology)

Level	Criterion
0	No shared conceptual apparatus. Groups use different terms for the same processes. Mutual incomprehension.
1	Terms are understood. Participants can read a document of the other side. But do not use the shared language in everyday speech.
2	Shared language is used in working documents. Meetings are conducted in the shared language. Terms do not require explanation.
3	Shared language is used for joint problem-solving. New terms are created jointly. Language adapts to new tasks.
4	Language is generated by the system independently. New concepts emerge organically, without external impulse. Language becomes an environment, not a tool.

15.4.2 Z₂. Agency (state)

Level	Criterion
0	No awareness of role. The element does not understand why it is in the system.
1	Execution by protocol. The element acts by instruction, does not initiate.
2	Initiative within the task. The element adapts actions to the situation.
3	Independent decision-making. The element acts as a subject, not an object.
4	Agenda-setting. The element generates directions, reproduces agency.

15.4.3 Z₃. Vision (dynamics)

Level	Criterion
0	No vision. The system does not formulate where it is going.
1	Vision is formulated by one center. Not shared. Transmitted from above.
2	Vision is discussed. Participants understand the direction but do not influence it.
3	Vision is shared. Participants adjust the direction jointly.
4	Vision is generated by the system. New directions emerge organically.

15.5 Axis Y — Quality

What the system creates and how it is embedded in the environment.

15.5.1 Y₁. Embeddedness (topology)

Level	Criterion
0	What is created does not exist in everyday life. A project on paper. No use.
1	Formal integration. Protocol is signed. Procedure exists. But is not used in everyday practice.

2	Use in everyday life. Participants turn to what is created without compulsion. Embedded in routine.
3	Self-reproduction. What is created is updated by participants without external impulse. The procedure lives, not merely exists.
4	Generation of new. Participants create new formats based on what is embedded. Embeddedness becomes a source of innovation.

15.5.2 Y₂. Adaptation (state)

Level	Criterion
0	No accounting for needs. What is created does not correspond to the environment.
1	Formal accounting. Needs are recorded but do not influence the result.
2	Substantive accounting. What is created is adapted to key needs.
3	Deep adaptation. What is created accounts for hidden needs, unformulated.
4	Joint adaptation. Environment and what is created adapt to each other.

15.5.3 Y₃. Innovation (dynamics)

Level	Criterion
0	No new formats. The system reproduces the existing.
1	New formats are proposed from outside. The system does not generate.
2	New formats emerge inside but do not take root.
3	New formats emerge and are embedded. The system renews.
4	New formats are generated continuously. Innovation becomes the norm.

15.6 Axis X — Connections

Structure of links between elements of the system.

15.6.1 X₁. Breadth (topology)

Level	Criterion
0	No channels. Elements are not connected.
1	One channel. Interaction through a single intermediary.
2	Several independent channels. Interaction does not depend on one intermediary.
3	Channels cover the majority of elements. The network is dense.
4	Channels are generated independently. New connections emerge without external impulse.

15.6.2 X₂. Depth (state)

Level	Criterion
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0	No interaction. Groups do not know each other. Channel is absent.
1	Interaction by protocol. Request — response. Formal. Outside the protocol, no contacts.
2	Consultation. One side helps the other formulate a task. Exchange is substantive, not formal.
3	Joint decision-making. Sides defend each other before third parties. The boundary “us — them” is blurred.
4	Exchange of strategic information. Joint risk management. Sides act as a single subject.

15.6.3 X₃. Density (dynamics)

Level	Criterion
0	No interaction. Channels exist but are not used.
1	Interaction is rare, on request. Reactive.
2	Interaction is regular. By schedule.
3	Interaction is continuous. Without schedule, by need.
4	Interaction generates new occasions. The flow creates itself.

15.7 Axis W — Resources

Realized connection through resources — volume, resilience, temporal structure.

15.7.1 W₁. Share (topology)

Level	Criterion
0	No exchange of resources.
1	Exchange through one center. All flows through one channel.
2	Exchange through several channels. Share of non-central channels grows.
3	Exchange is diversified. No dependence on one source.
4	Exchange generates new flows. Resources are attracted independently.

15.7.2 W₂. Resilience (state)

Level	Criterion
0	No exchange. No dependence. No resilience.
1	Exchange through one channel. Upon loss of the channel, exchange stops. The system does not withstand loss.
2	Exchange through several channels. Upon loss of one channel, exchange continues but with delay. The system adapts.
3	Exchange is resilient to loss of a key element. The system restructures without external intervention.

4	Exchange generates new channels independently. The system expands without external impulse. Resilience becomes a source of growth.
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15.7.3 W₃. Duration (dynamics)

Level	Criterion
0	No commitments. Interaction is one-time.
1	Short-term commitments. Horizon — one cycle.
2	Medium-term commitments. Horizon — several cycles.
3	Long-term commitments. Horizon — years. Joint planning.
4	Open-ended commitments. The system reproduces commitments independently.

15.8 Summary Table

Parameter	Description
Structure	12 dimensions × 5 levels = 60 cells
Cell content	Observable criterion
How to use	Find dimension → find level → check criterion
Principle	Do not interpret. Record.
Result	Level confirmed or not confirmed

16 Chapter 16. Journal and Monitoring

In Chapter 14 it was described how a snapshot is recorded. In Chapter 15 the instrument for determining the level was given. Here it is described how to work with snapshots over time. Journal and monitoring.

16.1 Journal

A journal is a sequence of snapshots recorded at equal intervals. Not a collection. Not an archive. A sequence. Order matters.

One snapshot shows state. Two snapshots show change. Three snapshots show direction. Four to five snapshots show a trajectory (Chapter 12). A journal is the material carrier of a trajectory. Without a journal, a trajectory does not exist. There are only memories. And memories are not data.

A journal contains for each snapshot: date, twelve levels, confirming events. Not interpretations. Not conclusions. Events. Conclusions are built later, on the basis of the journal. But the journal itself is only facts.

16.2 Monitoring

Monitoring is the regular recording of snapshots. Not continuous. Regular.

Continuous monitoring is impossible and not needed. The system does not change continuously. It changes discretely: an event occurred — a level changed. Between events, the level is stable. Recording it every day is meaningless. Every day it will be the same.

Regularity is determined by the cycle of the system (Chapter 12). Not arbitrarily. Not by convenience. By the speed of processes. If the system changes slowly — a snapshot once every several cycles. If quickly — once per cycle. Optimal frequency: one snapshot per cycle. Not more often. Not less often.

Monitoring that is too frequent gives noise: levels do not have time to change, but the snapshot already records. Monitoring that is too rare gives blindness: levels have changed, but the snapshot did not record. One snapshot per cycle is the balance between noise and blindness.

Two types of snapshots.

Scheduled snapshot. Recorded once per cycle. Shows trend. Answers the question: has anything changed over the cycle? If no — the system is stable. If yes — what exactly changed?

Event-driven snapshot. Recorded upon the fact of an event affecting the system. Without waiting for the next cycle. Shows a transition point. Answers the question: how did the event change the system?

An event affecting the system is not any event. It is an event that changes at least one dimension by at least one level. Departure of a key person. Merger. Crisis. Launch of a new direction. Change of regulatory requirements. If an event has occurred and there is reason to believe it affected the system — a snapshot is recorded.

A scheduled snapshot shows that the system is alive. An event-driven snapshot shows that the system has changed. Both are necessary. Without scheduled, there is no trend. Without event-driven, there is no transition point.

16.3 What to Record

For each dimension, the following is recorded: level and confirming event.

Level is a number from 0 to 4. Event is a fact that confirms this level. Not an opinion. Not a feeling. A fact.

Correct: " $X_2 = 2$. The buyer consulted the supplier on specification. Date: March 15."

Incorrect: " $X_2 = 2$. Trust is growing. Relations are improving. Positive dynamics."

The first is reproducible. A second observer, seeing the same event, will record the same level. The second is not reproducible. A second observer, seeing the same "relations," may record a different level.

A journal built on facts is data. A journal built on opinions is literature. The model works with data.

16.4 How to Read a Journal

A journal is read by three parameters: trends, anomalies, transition points.

16.4.1 Trends

How levels change from snapshot to snapshot. Grow, fall, stand still. A trend is a direction visible on a sequence of three or more snapshots. One snapshot does not show a trend. Two snapshots show change but not direction. Three snapshots show direction.

A trend is recorded at two levels.

Within axes. Spread between the three dimensions of each axis. If spread ≤ 1 — the axis is synchronized. The mean reflects reality. If spread ≥ 2 — the axis is desynchronized. The mean is misleading. The real level of the axis is determined by the minimum (Chapter 5). In the journal, desynchronization within an axis is marked with "!" after the mean. "2.33!" means: mean is 2.33, but within the axis spread ≥ 2 . Look at the minimum, not the mean.

Between axes. Spread between the means of the four axes. If spread ≤ 1 — the system is synchronized. If spread ≥ 2 — the system is desynchronized. Priority to the lagging axis.

Both trends are independent. The system can synchronize within axes and desynchronize between axes. And vice versa.

Trend by axis: the mean of three dimensions grows, falls, or is stable. Trend by system: the mean of four axes grows, falls, or is stable. Trend by spread: spread grows (desynchronization) or falls (synchronization).

16.4.2 Anomalies

Deviations from the trend. If three snapshots show growth and the fourth shows a fall, this is an anomaly. An anomaly requires explanation. Either a destructive event occurred. Or previous levels were inflated. Or the snapshot was recorded incorrectly.

An anomaly is not a diagnosis. An anomaly is a question. Why? What happened? The answer to the question is given by context, which is not in the journal. Context is in events, in people, in the environment. The journal records what. Context explains why.

16.4.3 Transition Points

Snapshots in which a level changed by 1 or more. A transition point is recorded by an event-driven snapshot — immediately, upon the fact of the event. A scheduled snapshot can record a transition after the fact: the level changed, but when and why is not visible. An event-driven snapshot records the transition at the moment it occurs. This allows reconstructing the mechanics: what triggered it, what followed, what lagged.

A transition point is recorded separately. For each transition point, the following is recorded: which dimension changed, from what level to what, what event confirms the transition. This allows later reconstructing the mechanics of the transition: what triggered it, what followed, what lagged.

16.5 Examples from Different Domains

16.5.1 Domain: product / user

Monitoring of a digital product. Cycle: month. Snapshot once a month. Journal for six months:

Snapshot	Z	Y	X	W	Spread between axes
January	1.33	1.00	1.33	0.33	1.00
February	1.67	1.33	1.67	0.33	1.34
March	2.00	1.67	2.00	0.67	1.33
April	2.33	2.00	2.33	0.67	1.66
May	2.33	2.00	2.33	1.00	1.33
June	2.67	2.33	2.67	1.00	1.67

Spread within axes: ≤ 1 on all snapshots. “!” is not placed. Means reflect reality.

Trend: growth on Z, Y, X. W lags but follows. Spread between axes slightly exceeds 1 in April and June due to W. This is a growth zone: W will follow.

Transition point: March. Z₁ transitioned from 1 to 2. Event: the team for the first time held a strategic session with users. Shared language is formed.

Anomaly: none. Trend is stable.

16.5.2 Domain: urban management

Monitoring of a city program. Cycle: year. Snapshot once a year. Journal for four years:

Snapshot	Z	Y	X	W	Spread between axes
Year 1	1.00	0.67	1.00	0.33	0.67
Year 2	1.67	1.33	1.67	0.67	1.00
Year 3	2.33	2.00	2.33	1.00	1.33
Year 4	2.33	2.00	2.33	1.00	1.33

Spread within axes: ≤ 1 on all snapshots. “!” is not placed.

Trend: growth in years 1–3. Plateau in year 4. Z, Y, X did not change. W did not change. Spread is stable.

Transition point: year 2. X_2 transitioned from 1 to 2. Event: residents for the first time jointly with the administration designed a public space. Trust became substantive.

Anomaly: year 4. Trend stopped. Three snapshots of growth, one snapshot of plateau. This can be a structural ceiling (Chapter 7) or a temporary pause. Diagnostics required: is effort being conducted? If yes, and there is no growth for three cycles — ceiling. If no — stagnation.

16.5.3 Domain: education / research school

Monitoring of a research group. Cycle: semester. Snapshot once a semester. Journal for three semesters:

Snapshot	Z	Y	X	W	Spread between axes
Semester 1	2.33	2.00	2.33	1.67	0.66
Semester 2	2.33	2.00	2.00	1.67	0.33
Semester 3	2.33!	1.67	1.67	1.67	0.66

Spread within axes: semesters 1–2 — ≤ 1 , “!” is not placed. Semester 3 — spread within Z = 2 ($Z_1 = 3$, $Z_2 = 1$, $Z_3 = 3$). “!” is placed. Mean 2.33 is misleading. The real level of Z is determined by the minimum: 1. Agency has fallen. Language and vision still hold. But the axis operates at level 1, not at level 2.33.

Trend: fall on Z, Y, X. W is stable. Spread between axes falls, but not due to synchronization of growth, but due to synchronization of fall.

Transition point: semester 3. Z_2 transitioned from 3 to 1. Event: the leader of the group went on sabbatical. Agency was not distributed. Cascade: Y and X began to fall.

Anomaly: W is stable while Z, Y, X are falling. This is a zone of false well-being (Chapter 11). Grants are approved, funding flows. But structure is degrading. If the trend continues, W will fall in 1–2 cycles. Action: urgent. Redistribution of agency. Do not wait for W to fall.

16.6 Summary Table

Parameter	Description
Journal	Sequence of snapshots at equal intervals
Monitoring	Regular recording, not continuous. One snapshot per cycle.
Types of snapshots	Scheduled (once per cycle) and event-driven (upon fact of event)
What to record	Level + confirming event. Not interpretation.
Trends	Direction on a sequence of 3+ snapshots. Two levels: within axes and between axes.
"!" mark	Desynchronization within axis ($\text{spread} \geq 2$). Real level is the minimum.
Anomalies	Deviation from trend. Requires explanation.
Transition points	Jump of level by 1+. Recorded by event-driven snapshot.

Type of snapshot	When recorded	What it shows
Scheduled	Once per cycle	Trend: has anything changed
Event-driven	Upon fact of event	Transition point: how the event changed the system

Summary of Part IV. Diagnostics

Part IV describes the instruments that turn the model into practice. Three instruments: snapshot, atlas, journal.

Chapter	Instrument	Essence
14	Snapshot	One-time recording of 12 dimensions. Observable events, not opinions. Presumption of unconfirmed.
15	Diagnostic atlas	60 cells: 12 dimensions $\times$ 5 levels. Observable criteria for each. Do not interpret. Record.
16	Journal and monitoring	Sequence of snapshots. Two types: scheduled (once per cycle) and event-driven (upon fact of event). Trends, anomalies, transition points. “!” mark — desynchronization within axis.

These instruments work in sequence. A snapshot records state. The atlas determines the level. The journal shows movement. Without a snapshot there is no data. Without the atlas there is no level. Without the journal there is no trajectory.

PART V

EXAMPLES

Summary of Part V. Examples

Part V shows how the language works in five domains. Each example: system → snapshot → diagnosis → trajectory → action. Not theory. Practice.

Example	Domain	Key diagnosis	Priority
1	M&A	Z and X lead Y. W lags. Growth zone.	Y. Quality is not embedded.
2	Urban management	All axes at level 0-1. Y at zero. Early stage.	Z. Identity first.
3	Startup / venture	Z leads Y, X, W. Spread = 1.67. Growth zone.	Y. Feedback does not influence product.
4	Architecture firm	Z lags. Agency in one person. Plateau.	Z. Distribution of agency.
5	Political party	All axes at level 0-1. Early stage.	Z. Language does not extend beyond activists.

Five domains. Five different configurations. One language. One set of rules. One structure of diagnosis.

17 Example 1. M&A

17.1 System

Two organizations after a merger deal. The goal is coupling: creating a single system from two previously independent ones. The system exists from the moment the deal closes. Before closing — two separate systems. After — one, in the process of formation.

Scale: two teams of 30–50 people each. Cycle: quarter. Observation horizon: two years (eight quarters).

17.2 Snapshot. Quarter 2

Snapshot recorded six months after the deal closed. Scheduled snapshot. No event-driven snapshots.

Dimension	Level	Confirming event
Z ₁ . Language	2	Teams use shared terminology in working documents. Recorded in three joint reports.
Z ₂ . Agency	1	Decisions are made by the leadership of the combined structure. Teams execute, do not initiate.
Z ₃ . Vision	2	A joint strategic session was held. Vision is formulated but not shared by all.
Y ₁ . Embeddedness	1	Processes are integrated formally. Protocol is signed. Everyday practice has not changed.
Y ₂ . Adaptation	1	Proposals of one team do not account for the specifics of the other. Standard solutions without customization.
Y ₃ . Innovation	0	No joint developments. No new formats have emerged.
X ₁ . Breadth	2	Three joint working groups have been created. Channels exist.
X ₂ . Depth	1	Interaction by protocol. Outside working groups, contacts are minimal.
X ₃ . Density	2	Meetings are weekly. Document exchange is regular.
W ₁ . Share	1	Budgets are consolidated, but distribution is through one center.
W ₂ . Resilience	1	Upon departure of a key manager of one of the teams, the process will stop.
W ₃ . Duration	2	Integration contract is signed for two years.

17.3 Diagnosis

Means: Z = 1.67, Y = 0.67, X = 1.67, W = 1.33.

Spread within axes: $Z = 1$, $Y = 1$, $X = 1$, $W = 1$. All axes are synchronized. “!” is not placed.

Spread between axes: $\max(Z, X) = 1.67$, $\min(Y) = 0.67$. Spread = 1.00. The system is synchronized.

Zone: growth. Z and X lead Y. W lags. Potential is accumulated on identity and connections. Quality is not embedded. Resources are not realized.

Priority: Y. Not Z, not X. Y. Quality is the lagging axis. Until quality is embedded, the system will not transition to the next level.

17.4 Trajectory. Quarters 2-5

Quarter	Z	Y	X	W	Spread between axes
2	1.67	0.67	1.67	1.33	1.00
3	2.00	1.00	2.00	1.33	1.00
4	2.33	1.33	2.33	1.67	1.00
5	2.33	1.67	2.33	1.67	0.66

Trend: growth on all axes. Y grows slower but grows. W follows with a delay of one quarter. Spread between axes narrows. The system synchronizes.

Transition point: quarter 3. Y_1 transitioned from 1 to 2. Event: the first joint working group adapted a process to the specifics of both teams. Quality is embedded in everyday life.

Anomaly: none. Trend is stable.

17.5 Action

Based on the trajectory: maintenance. Do not force. The system moves upward in concert. Y is catching up. W will follow.

Small steps: weekly meetings of working groups, exchange of reports, joint process reviews. Maintain the current level.

Event: a joint project that requires adaptation of quality to both teams. Transitions Y from level 1 to level 2. Not a small step. An event.

Monitoring: scheduled snapshot once per quarter. Event-driven snapshot upon departure of a key person, change of leadership, change of contract terms.

17.6 What the Language Shows

Without language: “Integration is going normally. Teams work together. Budgets are merged.”

With language: “Z and X at level 2. Y at level 0.67. W at level 1.33. Spread between axes = 1.00. Growth zone. Priority — Y. Quality is not embedded. Until Y grows, the system will not transition to level 3. W will follow with a delay. Do not force W. Work on Y.”

The difference: the first is description. The second is diagnosis with action.

18 Example 2. Urban Management

18.1 System

A city with a population of 50 thousand people. The city-forming enterprise is a metallurgical factory. The factory provides 60% of jobs and 40% of the local budget. The goal is transition from dependence on a single source to a diversified economy with distributed agency.

Scale: administration, factory, small business, residents, educational institutions. Cycle: year. Observation horizon: five years.

18.2 Snapshot. Year 1

Snapshot recorded at the start of the development program. Scheduled snapshot. Baseline point.

Dimension	Level	Confirming event
Z ₁ . Language	1	Administration and factory use different terms for the same processes. No joint documents.
Z ₂ . Agency	1	Decisions are made by administration and factory. Residents execute, do not initiate.
Z ₃ . Vision	1	Development strategy is written by administration. Not discussed with residents. Transmitted from above.
Y ₁ . Embeddedness	0	Development projects exist on paper. Not present in everyday life.
Y ₂ . Adaptation	0	Projects do not account for residents' needs. Standard solutions from a manual.
Y ₃ . Innovation	0	No new formats. The city reproduces the existing.
X ₁ . Breadth	1	One channel: administration — factory. Other groups are not connected.
X ₂ . Depth	1	Interaction by protocol. Outside formal meetings, no contacts.
X ₃ . Density	1	Meetings are rare, on request. Reactive interaction.
W ₁ . Share	1	All resources through the factory. One channel. One center.
W ₂ . Resilience	0	Upon production cuts by the factory, the city will not withstand. No reserves.
W ₃ . Duration	1	The factory funds social projects. Horizon — one year. Revised annually.

18.3 Diagnosis

Means: $Z = 1.00$, $Y = 0.00$, $X = 1.00$, $W = 0.67$.

Spread within axes: $Z = 0$, $Y = 0$, $X = 0$, $W = 1$. All axes are synchronized. "!" is not placed.

Spread between axes: $\max(Z, X) = 1.00$, $\min(Y) = 0.00$. Spread = 1.00. The system is synchronized.

Zone: early stage. All axes at level 0-1. Y at zero. Quality is absent. Potential is not accumulated.

Priority: Z. Not Y, not X. Z. Identity is first. Without shared language, without distributed agency, without shared vision — quality cannot be created, connections cannot be built. Pattern $Z \rightarrow Y/X \rightarrow W$ (Chapter 10).

18.4 Trajectory. Years 1-5

Year	Z	Y	X	W	Spread between axes
1	1.00	0.00	1.00	0.67	1.00
2	1.67	0.67	1.67	0.67	1.00
3	2.33	1.33	2.00	1.00	1.33
4	2.33	2.00	2.33	1.00	1.33
5	2.67	2.33	2.67	1.33	1.34

Trend: growth on all axes. Z grows first. Y and X follow. W lags with a delay of one to two years. Spread between axes slightly exceeds 1 in years 3-5 due to W. Growth zone.

Transition point: year 2. Z_2 transitioned from 1 to 2. Event: residents for the first time initiated a landscaping project and received funding from administration without factory intermediation. Agency is distributed.

Transition point: year 3. X_2 transitioned from 1 to 2. Event: residents jointly with administration designed a public space. Trust became substantive.

Anomaly: none. Trend is stable. W lags, but this is the norm: resources diversify when the system becomes a subject.

18.5 Action

Based on the trajectory: continuation of work on Z, Y, X. Do not force W.

Small steps: annual city forums, joint working groups, information mailings. Maintain the current level.

Events: joint design of public spaces (transitions X_2), grant for resident initiative without conditions (transitions Z_2), joint project of administration and small business (transitions Y_1).

Monitoring: scheduled snapshot once per year. Event-driven snapshot upon change of factory leadership, change of federal policy, crisis.

Structural ceiling: W_1 . A city with a population of 50 thousand will not become a federal center of attraction. Share of external resources is limited by scale. This is not stagnation. This is a ceiling. Strategy: expansion — tourism, small business, creative industries. W_1 will not grow, but the overall volume of resources will increase through new channels.

18.6 What the Language Shows

Without language: “The city is developing. The factory helps. Residents participate. Projects are implemented.”

With language: “ $Z = 2.67$, $Y = 2.33$, $X = 2.67$, $W = 1.33$. Spread between axes = 1.34. Growth zone. W lags — norm. Do not force. Ceiling on W_1 — scale of the city. Strategy: expansion. Small steps: forums, working groups. Events: joint design, grants without conditions. Monitoring: once per year. Event-driven snapshot upon change of factory leadership.”

The difference: the first is description. The second is diagnosis with action, strategy, and monitoring.

19 Example 3. Startup / Venture

19.1 System

A startup at pre-seed stage. Team: three founders, two developers. Product: a digital platform for managing local services. The goal is transition from hypothesis to a stable model with paying users.

Scale: five people, first users, one investor. Cycle: month. Observation horizon: twelve months.

19.2 Snapshot. Month 3

Snapshot recorded three months after the start of work. Scheduled snapshot. Product in closed testing.

Dimension	Level	Confirming event
Z ₁ . Language	2	The team uses shared terminology. Users understand the interface without explanation. Recorded in five testing sessions.
Z ₂ . Agency	2	Founders distribute tasks independently. Decisions are made jointly. One founder adapts the product without approval.
Z ₃ . Vision	2	Vision is formulated. The team understands the direction. But vision is not discussed with users.
Y ₁ . Embeddedness	1	The product is used by a test group. But not embedded in everyday life. Used on tester request.
Y ₂ . Adaptation	1	Feedback is collected but does not influence the product. Features are implemented according to founders' plan.
Y ₃ . Innovation	1	One new format proposed by a user. Not implemented.
X ₁ . Breadth	1	One channel: founders — test group. Other segments are not covered.
X ₂ . Depth	1	Interaction by testing protocol. Outside sessions, no contacts.
X ₃ . Density	2	Testing sessions are weekly. Feedback is regular.
W ₁ . Share	0	No revenue. Product is free. Investment is a single source.
W ₂ . Resilience	0	Upon cessation of investment, the team will not survive. No reserves.
W ₃ . Duration	1	Investment for six months. Horizon — one funding cycle.

19.3 Diagnosis

Means: Z = 2.00, Y = 1.00, X = 1.33, W = 0.33.

Spread within axes: $Z = 0$, $Y = 0$, $X = 1$, $W = 1$. All axes are synchronized. “!” is not placed.

Spread between axes: $\max(Z) = 2.00$, $\min(W) = 0.33$. Spread = 1.67. The system is desynchronized.

Zone: growth. Z leads Y, X, W. Identity is formed. Quality is not embedded. Connections are formal. Resources are absent.

Priority: Y. Not X, not W. Y. Quality is the lagging axis. The product exists but is not adapted to users. Feedback is collected but does not influence the result. Until Y grows, X will not deepen, W will not appear.

Desynchronization between axes (spread = 1.67) is not a problem. This is early stage. Z transitions first (Chapter 10). Y and X will catch up. W will follow last. But the spread must be monitored. If it grows to 2 — desynchronization becomes critical.

19.4 Trajectory. Months 3-9

Month	Z	Y	X	W	Spread between axes
3	2.00	1.00	1.33	0.33	1.67
4	2.33	1.33	1.67	0.33	2.00
5	2.33	1.67	2.00	0.33	2.00
6	2.67	2.00	2.33	0.67	2.00
7	2.67	2.33	2.67	1.00	1.67
8	3.00	2.67	3.00	1.00	2.00
9	3.00	3.00	3.00	1.33	1.67

Trend: growth on all axes. Z grows first. Y and X follow with a delay of one month. W lags with a delay of three months. Spread between axes reaches 2.00 in months 4-6 and 8. Critical zone. But by month 9 the spread narrows to 1.67. The system synchronizes.

Transition point: month 5. Y_2 transitioned from 1 to 2. Event: the team for the first time adapted a feature based on user feedback. Quality became substantive.

Transition point: month 7. X_2 transitioned from 2 to 3. Event: a user independently proposed an improvement that the team accepted without corrections. Trust became joint.

Transition point: month 8. Z_1 transitioned from 2 to 3. Event: users began using the product’s terminology in communication with each other. Language became an environment.

Anomaly: month 4. Spread between axes reached 2.00. Z grew, Y and X did not keep up. W at zero. This is not an anomaly. This is a risk zone: Z leads Y and X. If Y does not catch up within two months, the gap becomes critical. Action: priority Y. Not Z. Not X. Y.

19.5 Action

Based on the trajectory: work on Y. Do not force W. Do not build up Z.

Small steps: weekly testing sessions, feedback collection, exchange of reports within the team. Maintain the current level.

Events: adaptation of a feature based on feedback (transitions Y_2), acceptance of a user improvement without corrections (transitions X_2), joint strategic session with users (transitions Z_3).

Monitoring: scheduled snapshot once per month. Event-driven snapshot upon receiving investment, departure of a founder, launch of a competitor, change of regulatory requirements.

Structural ceiling: W_1 at pre-seed stage. Revenue will not appear without paying users. Paying users will not appear without embedded quality. Quality is not embedded. The ceiling on W is a consequence, not a cause. The cause is Y. Work on Y. W will follow.

19.6 What the Language Shows

Without language: “The startup is developing. The team works. Users test. The investor is satisfied.”

With language: “ $Z = 2.00$, $Y = 1.00$, $X = 1.33$, $W = 0.33$. Spread between axes = 1.67. Growth zone. Priority — Y. Feedback is collected but does not influence the product. Until Y grows, W will not appear. Spread will reach 2.00 in month 4 — risk zone. Do not force Z. Work on Y. Monitoring: once per month. Event-driven snapshot upon receiving investment or departure of a founder.”

The difference: the first is description. The second is diagnosis with priority, spread forecast, and action.

20 Example 4. Architecture Firm

20.1 System

An architecture firm in a regional center. Seven architects, two partners, chief architect. Specialization: public buildings, urban environment. The goal is transition from executing orders to shaping the architectural agenda of the region.

Scale: nine people, 5–8 projects per year, three permanent clients. Cycle: year. Observation horizon: four years.

20.2 Snapshot. Year 1

Snapshot recorded at the start of observation. Scheduled snapshot. The firm has been operating for five years. Market position is stable but not growing.

Dimension	Level	Confirming event
Z ₁ . Language	2	The firm has a recognizable style. Clients understand the firm's terminology. But language does not extend beyond the firm.
Z ₂ . Agency	1	Decisions are made by the chief architect. Architects execute. Partners approve, do not initiate.
Z ₃ . Vision	1	Vision is formulated by the chief architect. Not discussed with the team. Transmitted from above.
Y ₁ . Embeddedness	2	The firm's projects are realized. Built objects are used. The firm is present in the urban environment.
Y ₂ . Adaptation	2	Projects are adapted to clients' needs. Customization exists. But hidden needs are not accounted for.
Y ₃ . Innovation	1	New formats are proposed by the chief architect. The team does not generate. One experimental project in three years.
X ₁ . Breadth	2	Three permanent clients, connections with contractors, participation in professional conferences.
X ₂ . Depth	2	Clients consult at early stages. Exchange is substantive. But no joint decisions.
X ₃ . Density	2	Interaction is regular. By project cycle. Not continuous but stable.
W ₁ . Share	2	Three permanent clients provide 70% of revenue. Two new clients in the last year.
W ₂ . Resilience	1	Upon departure of the chief architect, the firm will lose style and clients. Dependence on one person.
W ₃ . Duration	2	Contracts for two to three years. Joint planning with two of three clients.

20.3 Diagnosis

Means: Z = 1.33, Y = 1.67, X = 2.00, W = 1.67.

Spread within axes: $Z = 1$, $Y = 1$, $X = 0$, $W = 1$. All axes are synchronized. “!” is not placed.

Spread between axes: $\max(X) = 2.00$, $\min(Z) = 1.33$. Spread = 0.67. The system is synchronized.

Zone: plateau. The firm is stable. Levels 1-2. No growth. Quality and connections at level 2. Identity lags: agency is concentrated in one person, vision is not shared.

Priority: Z. Not Y, not X. Z. Identity is the lagging axis. Until agency is distributed and vision is shared, the firm will not transition to level 3. Quality and connections hold at level 2 but do not grow because Z does not allow it.

20.4 Trajectory. Years 1-4

Year	Z	Y	X	W	Spread between axes
1	1.33	1.67	2.00	1.67	0.67
2	1.33	1.67	2.00	1.67	0.67
3	2.00	2.00	2.33	2.00	0.33
4	2.33	2.33	2.67	2.33	0.34

Trend: stagnation in years 1-2. Growth in years 3-4. Spread between axes narrows. The system synchronizes.

Transition point: year 3. Z_2 transitioned from 1 to 2. Event: two architects for the first time independently led a project from concept to delivery, without participation of the chief architect. Agency is distributed.

Anomaly: years 1-2. Stagnation. Levels do not change for two years. Effort is conducted: the firm participates in conferences, maintains relations with clients. But no growth. Diagnosis: structural ceiling on Z_2 . The chief architect does not transfer agency. Vision is not discussed. The firm rests on one person.

Action in year 2: not expansion. Not alignment on Y or X. Work on Z. Event: strategic session with participation of the entire team. Vision is discussed. Agency is distributed. This is not a small step. This is an event. Without it, stagnation continues.

20.5 Action

Based on the trajectory: work on Z in years 1-2. Maintenance of Y and X. Do not force W.

Small steps: regular internal project reviews, participation in conferences, exchange of portfolio with clients. Maintain the current level.

Events: team strategic session (transitions Z_3), independent project leadership by architects without the chief (transitions Z_2), joint project with another firm (transitions X_1 and Y_3).

Monitoring: scheduled snapshot once per year. Event-driven snapshot upon departure of the chief architect, loss of a key client, receipt of a large order.

Structural ceiling: W_1 . A regional center limits scale. The firm will not become a federal player without relocation or opening a branch. This is not stagnation. This is a ceiling. Strategy: expansion — competitions, publications, teaching. W_1 will not grow, but reputation and influence will increase through new channels.

20.6 What the Language Shows

Without language: “The firm works stably. Clients are satisfied. Projects are high-quality. But no growth.”

With language: “ $Z = 1.33$, $Y = 1.67$, $X = 2.00$, $W = 1.67$. Spread between axes = 0.67. Plateau zone. Priority — Z. Agency is concentrated in the chief architect. Vision is not shared. Until Z grows, Y and X will not transition to level 3. Stagnation in years 1-2 — structural ceiling on Z_2 . Event: strategic session, distribution of agency. Monitoring: once per year. Event-driven snapshot upon departure of the chief architect.”

The difference: the first is description. The second is diagnosis with the cause of stagnation, priority, and a specific event.

CLOSING

21 What the Language Gives

The language gives a shared conceptual apparatus. Twelve dimensions. Four axes. Three aspects. Five levels. Spread. Synchronization. Phase. Transition. Terms that mean the same thing for all participants. Not “I feel that trust has grown.” But “X₂ transitioned from 1 to 2. Event: the buyer consulted on specification.”

The language gives diagnostics. A snapshot records state. The atlas determines the level. The journal shows movement. A trajectory shows direction. Spread shows desynchronization. The “!” mark shows where the mean is misleading. All of this is data. Not opinions. Data.

The language gives priorities. Upon desynchronization within an axis — priority to the lagging dimension. Upon desynchronization between axes — priority to the lagging axis. In a growth zone — do not force W. In a risk zone — work on the lagging. At a ceiling — alignment or expansion. Priority is determined by structure, not by intuition.

The language gives monitoring. Scheduled snapshot once per cycle. Event-driven snapshot upon the fact of an event. Trends. Anomalies. Transition points. The system is not observable continuously. It is observable discretely. And this is sufficient.

The language gives strategy. Small steps maintain. Events transition. A catalyst triggers a transition in a pre-critical system. A freeze fixes. A managed downgrade adapts. Closure completes. Each action has a condition and a result. Not a recommendation. A consequence of structure.

22 What the Language Does Not Give

The language does not give ready-made solutions. It shows where the system is. It does not show what to do. Action is determined by context: people, environment, resources, history. The language records what. Context explains why. The decision is made by a person.

The language does not give forecasts. A trajectory shows direction. It does not show what will happen. A forecast requires knowledge of future events. The language requires only knowledge of past states. The difference is fundamental.

The language does not give guarantees. The system can grow. It can degrade. It can collapse. The language shows probability, not inevitability. Level 3 does not guarantee stability. It gives time. If the system uses time for adaptation — stability is maintained. If it does not — degradation.

The language does not give interpretations. A snapshot records events. Not meanings. Meaning is assigned later, when building a diagnosis. At the stage of recording, meaning is not needed. Only this is needed: did the event occur or did it not. Two observers recording the same events will obtain the same snapshot. Two observers interpreting the same events will obtain different snapshots. The language is built on the first, not the second.

The language does not replace the expert. It structures observation. It does not replace judgment. The expert sees what the language does not record: intonation, context, history, hidden motives. The language records what the expert can miss: spread, desynchronization, lag of W. The language and the expert complement each other. They do not replace.

23 Invitation

The language is an instrument. Not a dogma. Not a theory. Not an ideology. An instrument.

An instrument is used. Not discussed. Not defended. Not propagated. Used.

Take a system. Any. An organization. A city. A product. A team. A family. A party. A firm. A school. A hospital. Make a snapshot. Twelve dimensions. Levels from 0 to 4. Confirming events. Not opinions. Events.

Look at the spread. Within axes. Between axes. Where is desynchronization? Where is the lagging? Where is the “!” mark?

Build a trajectory. Three snapshots. Four. Five. Where is the system moving? Up? Down? Sideways? Does spread grow or fall?

Determine priority. Not by intuition. By structure. The lagging element. The weak link. The risk zone. The ceiling.

Act. Small steps. Events. Monitoring. Snapshot. Another snapshot. Another.

Do not believe the language. Verify. If the snapshot shows one thing and reality shows another, the problem is not in reality. The problem is in the snapshot. Re-record. Clarify. Record more precisely.

The language is not truth. The language is an instrument of observation. Like a microscope. Like a ruler. Like a thermometer. A microscope does not create a cell. A ruler does not create length. A thermometer does not create temperature. They record. The language records.

Use it.

24 Glossary

Term	Definition	Chapter
Adaptation	Ability of what is created to account for environment needs. Y_2 .	2
Agency	Ability of an element to act as a subject. Z_2 .	2
Alignment	Strategy at ceiling: pulling lagging axes up to ceiling level.	7
Anomaly	Deviation from trend on a trajectory. Requires explanation.	16
Antifragility	Property of a system to be strengthened by stress. Stress transitions a ready system.	9
Aspect	Angle of view on a dimension: topology, state, dynamics.	3
Axis	Group of three dimensions: Z, Y, X, W.	2
Breadth	Number of independent channels. X_1 .	2
Cascade	Fall of one dimension pulls others. Mechanics of reverse transition.	11
Catalyst	Small step in a pre-critical system that triggers a transition.	8
Closure	Deliberate cessation of a system's existence.	13
Coupling	Transition of several axes simultaneously. Changes configuration.	10
Cycle	Period between snapshots. Determined by speed of processes.	12
Density	Frequency and continuity of interaction. X_3 .	2
Depth	Quality of interaction between elements. X_2 .	2
Desynchronization	Spread ≥ 2 . System loses coherence.	5
Diagnostic atlas	Table of observable criteria: 12 dimensions $\times$ 5 levels = 60 cells.	15
Duration	Horizon of commitments. W_3 .	2
Embeddedness	Degree of presence of what is created in everyday life. Y_1 .	2
Entropy	Tendency of a system to degrade without inflow of energy.	9

Event	Intervention that transitions the system to the next level.	8
Event-driven snapshot	Snapshot recorded upon the fact of an event. Shows transition point.	16
Expansion	Strategy at ceiling: entry into an adjacent domain.	7
Field separation	Z, Y, X — action field. W — result field. Do not manage W directly.	6
Freeze	Fixation of the system at the current level without attempts at growth.	13
Growth zone	Z, Y, X grow, W lags. Potential is accumulated.	6
Identity	Axis Z. Language, agency, vision.	2
Innovation	Ability of a system to generate new formats. Y ₃ .	2
Journal	Sequence of snapshots recorded at equal intervals.	16
Language	Shared conceptual apparatus. Z ₁ . Also: name of the model.	2
Managed downgrade	Deliberate transition of the system to a lower level.	13
Monitoring	Regular recording of snapshots. Scheduled and event-driven.	16
Norm	Acceptable spread. Within axis ≤ 1 . Between axes ≤ 1 .	5
Observable event	Fact recorded by an external observer. Basis of a snapshot.	14
Phase	Discrete state of a system. Levels 0–4.	4
Phase transition	Jump of a system from one phase to another.	10
Plateau zone	Levels do not change for three or more cycles.	12
Presumption of unconfirmed	No event — no level. Burden of proof on the one who asserts.	14
Quality	Axis Y. Embeddedness, adaptation, innovation.	2
Resources	Axis W. Share, resilience, duration.	2
Resilience	Ability of exchange to withstand loss of channels. W ₂ .	2
Reverse phase transition	Jump of a system from a higher phase to a lower one.	11

Risk zone	W grows while Z, Y, X lag. Exchange without structure.	6
Scheduled snapshot	Snapshot recorded once per cycle. Shows trend.	16
Share	Structure of resource distribution. W_1 .	2
Small step	Intervention that maintains the system at the current level. Does not transition.	8
Snapshot	One-time recording of 12 dimensions.	14
Spread	Difference between maximum and minimum level.	5
Stagnation zone	Levels do not change for three or more cycles.	12
State	Aspect: what quality an element possesses.	3
Structural ceiling	Objective constraint of the environment. Three cycles without shift — ceiling.	7
Synchronization	$\text{Spread} \leq 1$. System is coherent.	5
To transition	To change the phase of a system. Not “to translate.”	8
Topology	Aspect: how an element is structured.	3
Trajectory	Sequence of snapshots over time. Shows movement.	12
Transition point	Snapshot in which a level changed by 1+.	16
Trend	Direction on a sequence of 3+ snapshots.	16
Vision	Shared view of direction. Z_3 .	2
Weak link rule	The lagging element determines the real level of the system.	1
Zone of false well-being	W holds while Z, Y, X fall. Resources exist, structure is destroyed.	11